Lecture_3

CMOS 制造工艺

Manufacturing Process

参考：讲义 第3章

2020年09月29日

第3章 CMOS制造工艺简介
CMOS Manufacturing Process

- MOSFET结构及CMOS工艺层
- CMOS制造工艺
- 设计规则（Design Rule）
- IC封装（Packaging）

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- **MOSFET结构及CMOS工艺层**
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- **设计规则（Design Rule）**
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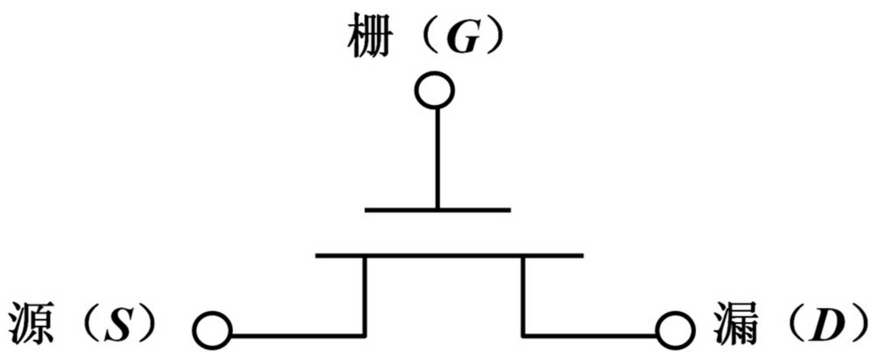
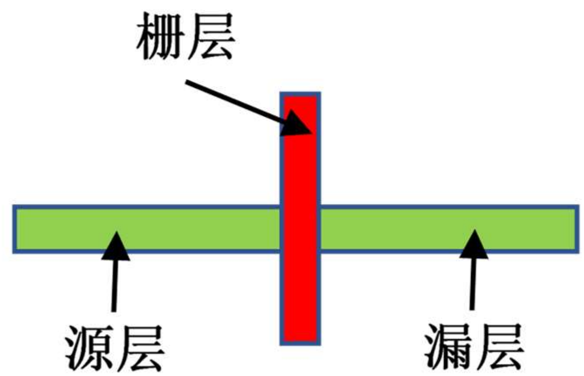
MOSFET结构

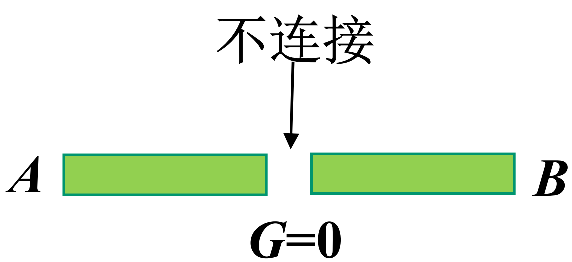
NMOS



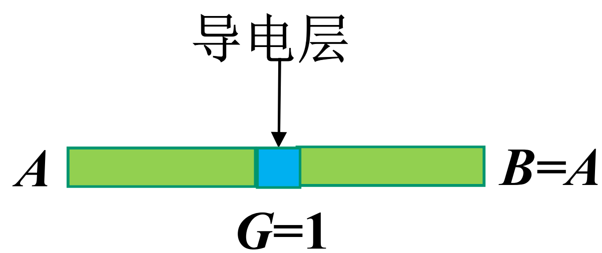
电路符号



工艺层



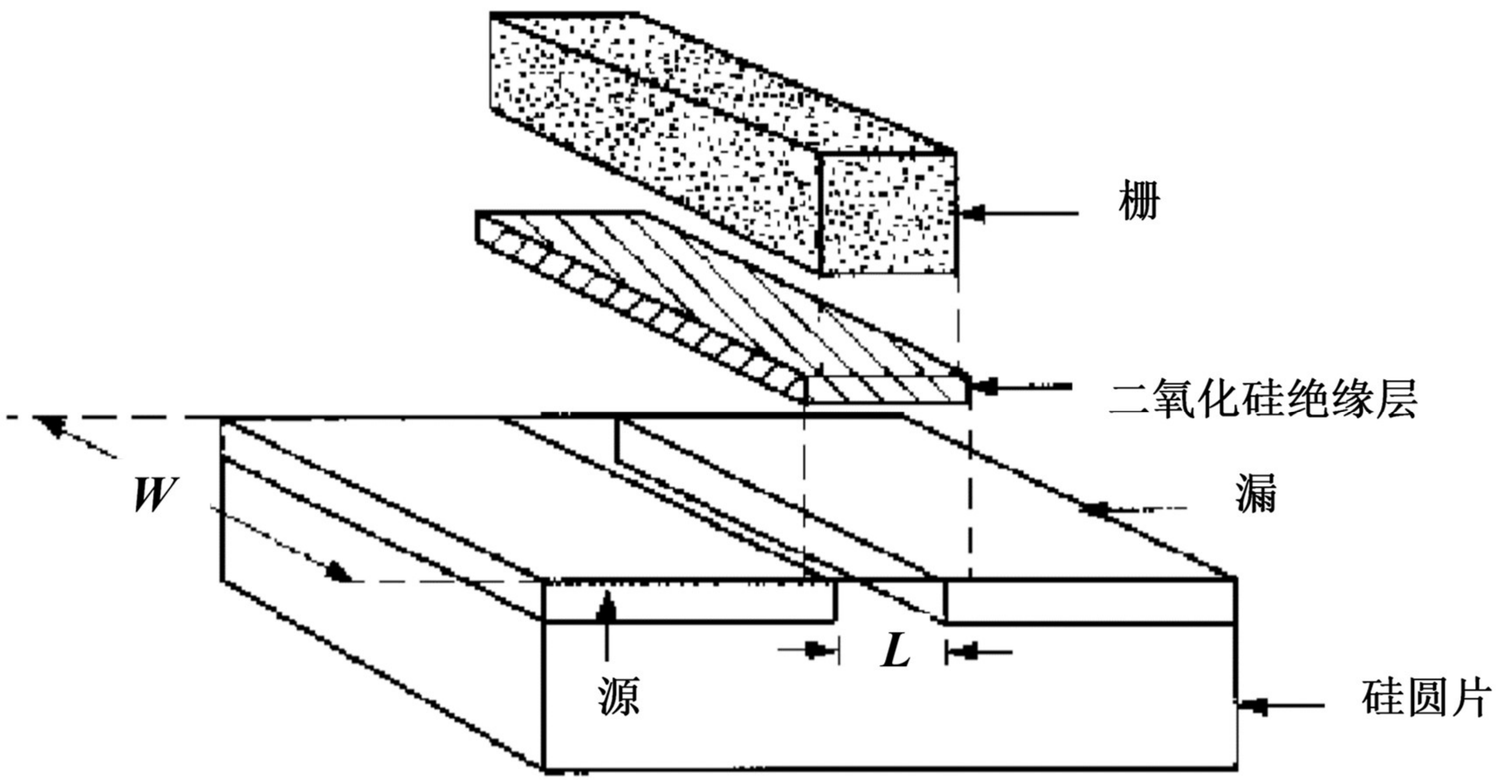
开关开路



开关闭合

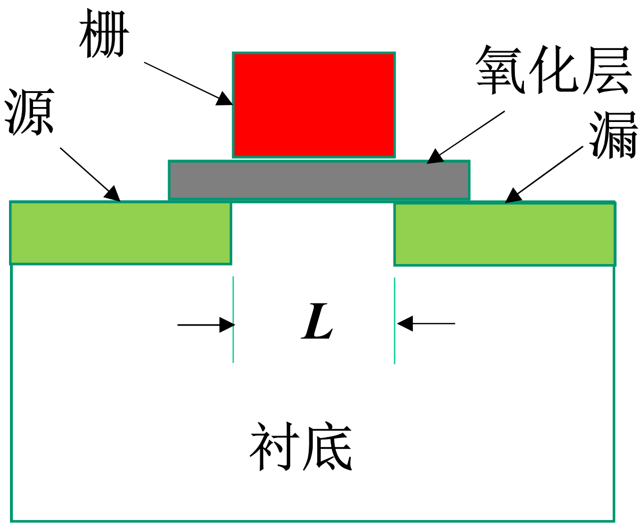
MOSFET结构

形成MOSFET的各工艺层

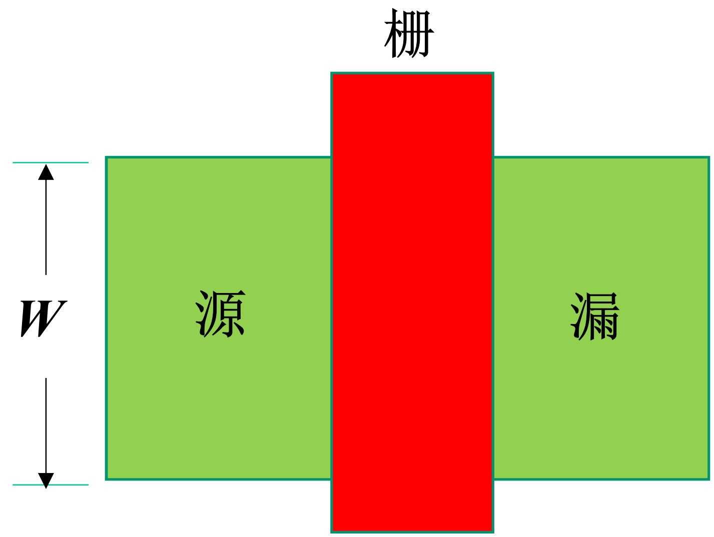


MOSFET结构

MOSFET视图



断面图



顶视图

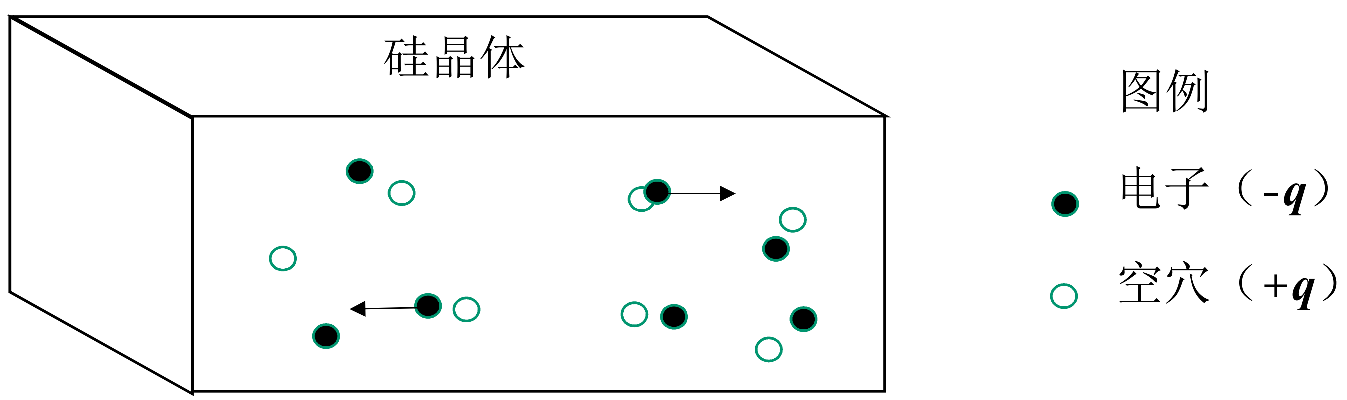
MOSFET结构

硅的导电性

硅晶体的原子密度约为$N_{Si} \approx 5 \times 10^{22}$个原子/cm³，但是其中只有很少量的电子能够导电

每立方厘米材料中可自由载流的电子数目用符号n_i表示，称为本征载流子密度

本征载流子密度：$n_i \approx 1.45 \times 10^{10}\ \mathrm{cm^{-3}}$



在一个纯净硅样品中有：$n = p = n_i$，n和p分别为每立方厘米中自由电子和自由空穴的数目。两者相乘得到：$n\,p = n_i^2$

MOSFET结构

硅的导电性

纯净硅是电的不良导体，但可通过有目的地加入少量的杂质原子（称为掺杂剂）来改变而形成掺杂样品，从而增加电子或空穴的数量以增加其导电性

在硅晶体中加入砷（**As**）或磷（**P**）原子可以增加自由电子的数量，得到的样品称为**n**型材料，因为它含有过剩的带负电荷的电子。砷和磷都能为晶体提供自由电子，所以被称为施主原子，或简称施主。每立方厘米中加入施主的数量用符号N_D表示

与**n**型极性相反的材料叫做**p**型材料，是通过向硅晶体中加入硼（**B**）原子形成的。**p**型材料中带正电荷的空穴数多于带负电荷的电子数。每个硼原子都会在化学键中形成一个自由空穴。由于一个空穴可以接受一个电子，硼被称为受主掺杂剂。每一立方厘米中加入的受主数量用符号N_A表示

MOSFET结构

硅的导电性

n型材料中，每个施主原子为晶体增加一个自由电子，所以电子密度为：

$$n_{\mathrm{n}} \approx N_{\mathrm{D}} \ \ \mathrm{cm}^{-3}$$

空穴的数量用符号p_{n}表示，

$$p_n = \frac{n_i^2}{N_D} \ \ \mathrm{cm}^{-3}$$

由于电子和空穴的相对数量不同，在**n**型材料中，电子被称为多子，而空穴则称为少子

p型材料中载流子的密度：

$$p_{\mathrm{p}} \approx N_{\mathrm{A}} \ , \qquad n_{\mathrm{p}} = \frac{n_i^2}{N_A}$$

同时由于$p_{\mathrm{p}} >> n_{\mathrm{p}}$，所以称空穴为多子载体，电子为少子载体

MOSFET结构

硅的导电性

一个载流子密度为n和p的半导体区域其电导率σ为:

$$\sigma = q\,(\mu_n\, n + \mu_p\, p)$$

μ_n和μ_p分别称为电子和空穴的迁移率，单位为$\mathrm{cm^2/(V\cdot s)}$

迁移率是说明一个粒子“如何移动”的参数：μ值小说明粒子移动困难，μ值大则说明移动相对自由。室温下本征硅的迁移率为：

$$\mu_n = 1360\ \mathrm{cm^2/(V\cdot s)} \quad \mu_p = 480\ \mathrm{cm^2/(V\cdot s)}$$

对于n型样品，若$n_n \gg p_n$，则其电导率可以近似为：$\sigma = q\,\mu_n\, n_n$

同样，p型区的电导率可以估计为：$\sigma = q\,\mu_p\, p_p$

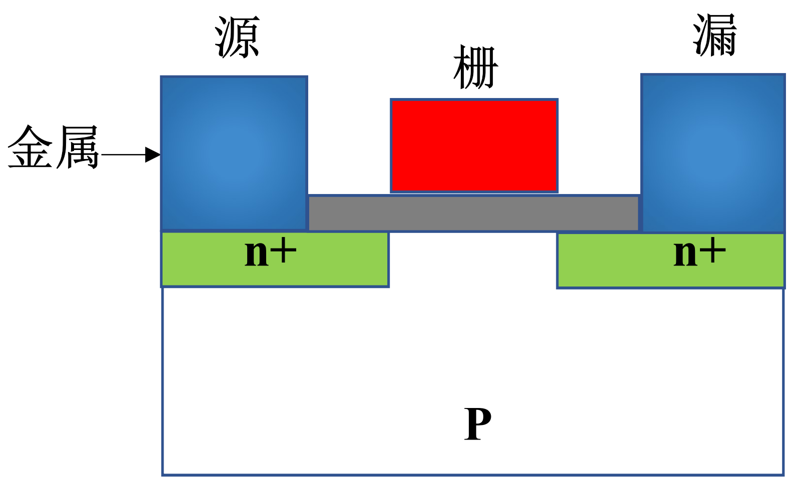
在n型区以带负电荷的电子为主，而在p型区则大多数为带正电荷的空穴

MOSFET结构

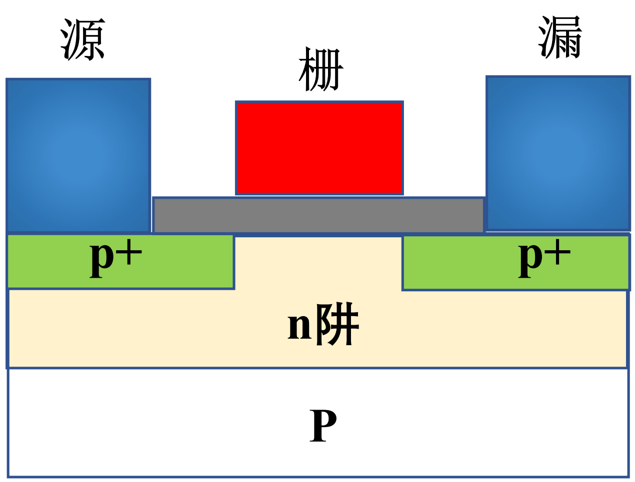
NMOS和PMOS结构

一个MOSFET的极性（n或p）是由漏区和源区的极性决定的。当器件导电时，导电层的极性与漏区和源区的极性相同

NMOS采用n型的漏区和源区，而PMOS则具有p型的漏区和源区



NMOS截面图

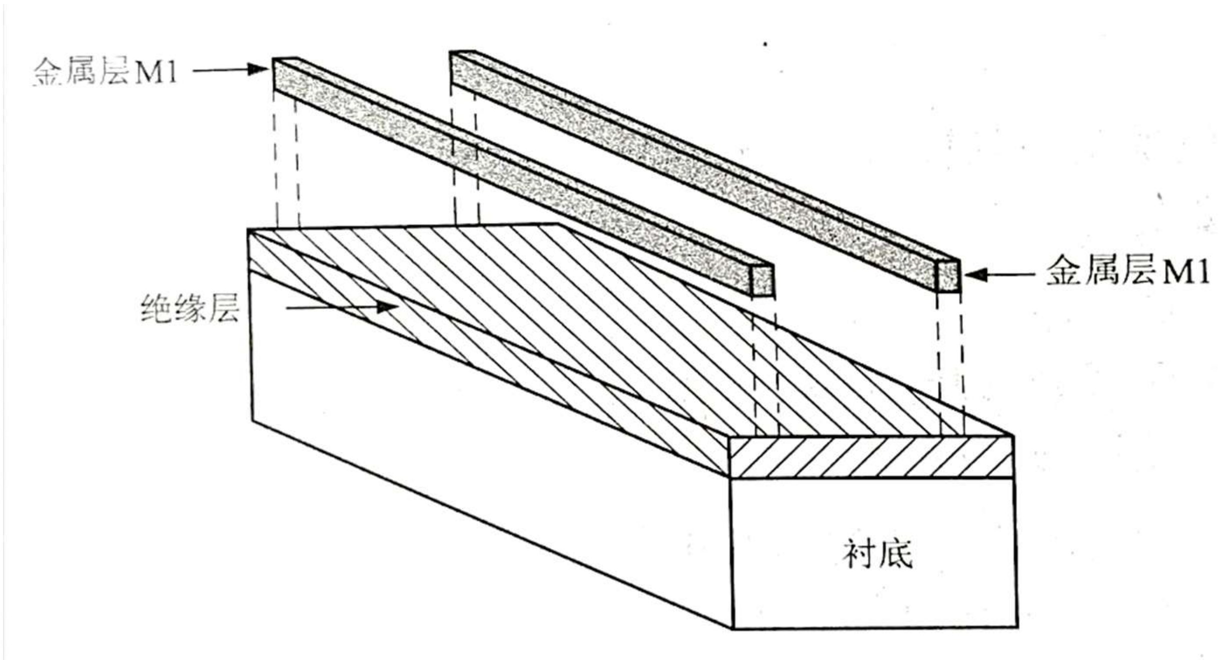


PMOS截面图

CMOS工艺层

集成电路工艺层概念

一个硅集成电路可以看成是一组形成图形的材料层的集合，它是由按特定次序将不同材料层叠在一起形成三维结构构成的，这些结构共同作用成为一个电子开关电路



两种不同的材料层

CMOS工艺层

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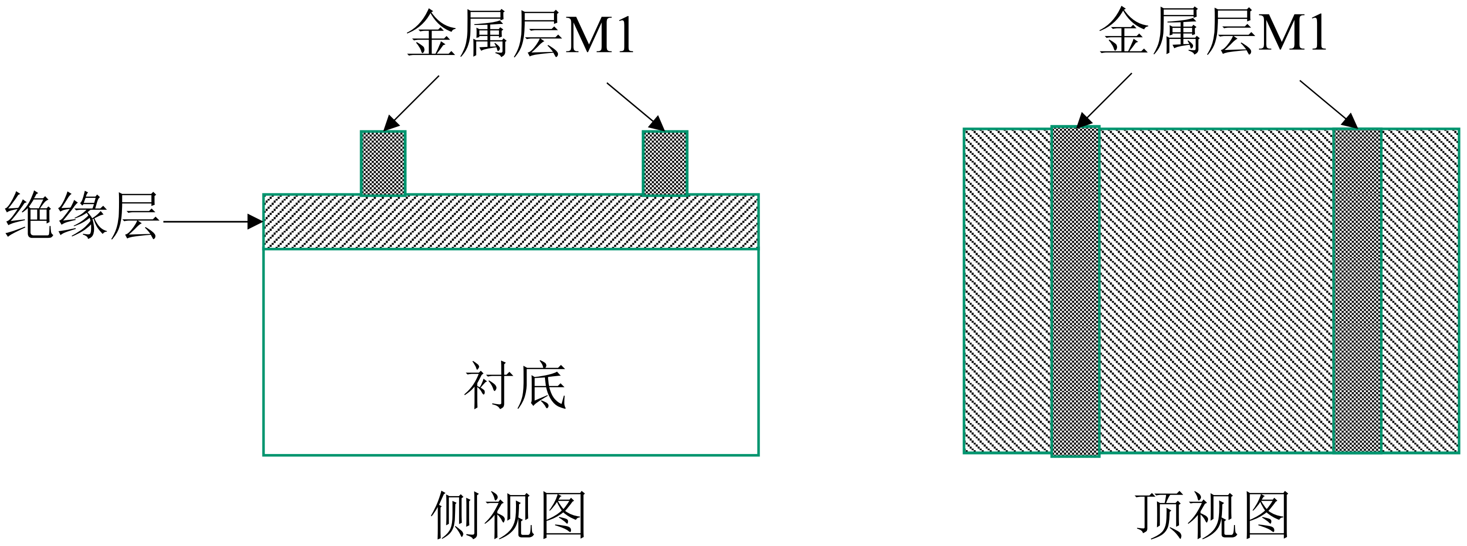

叠放过程完成后的各层情形

CMOS工艺层

集成电路工艺层概念

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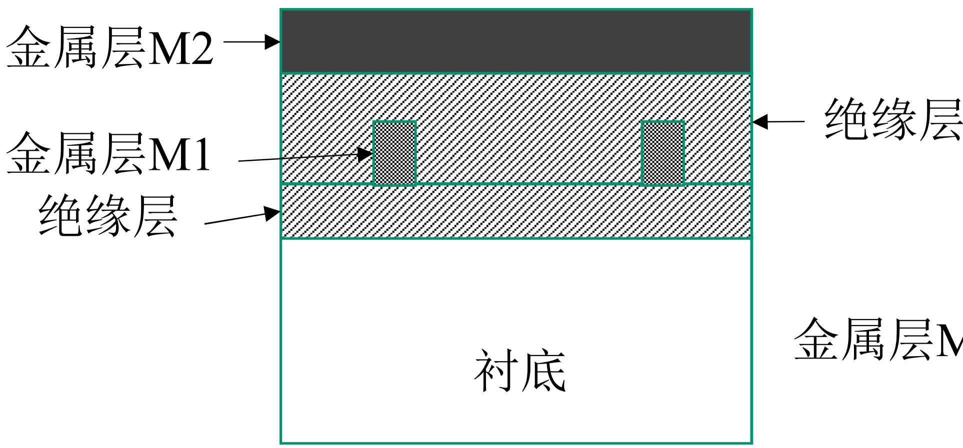
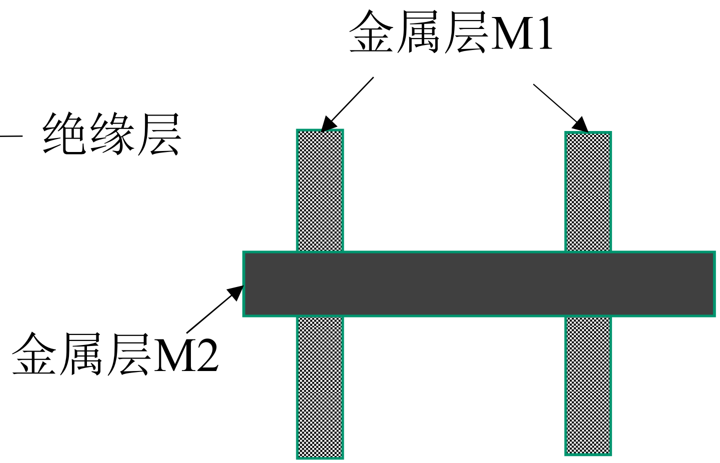


侧视图



顶视图

加上另一层绝缘层和第二层金属层

CMOS工艺层

集成电路工艺层概念

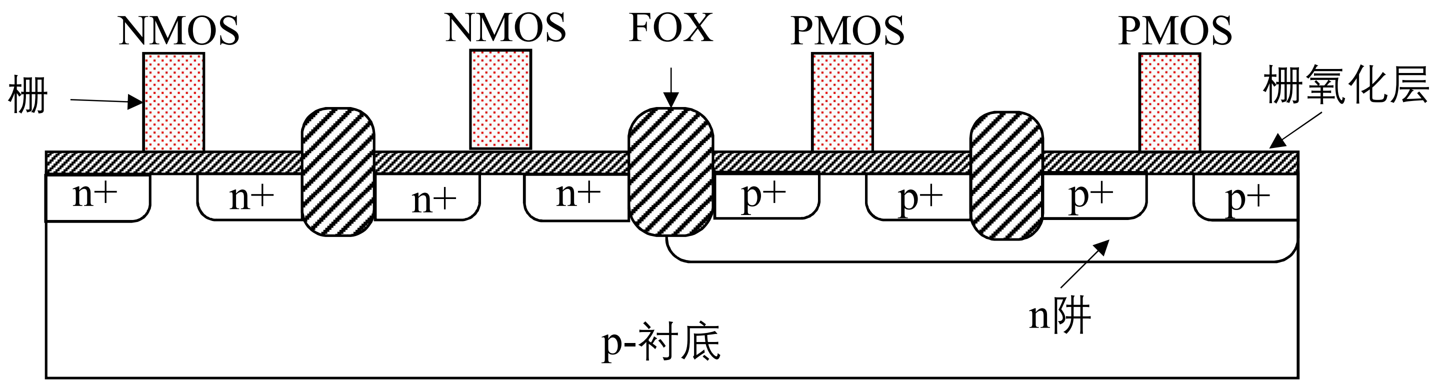
一个硅集成电路可以看成是一组形成图形的材料层的集合，它是由按特定次序将不同材料层叠在一起形成三维结构构成的，这些结构共同作用成为一个电子开关电路

将顶视图和侧视图结合在一起，可以看到集成电路的三维结构：

- 侧视图显示各层的叠放顺序
- 绝缘层将各图形层（如两金属层）分隔开，它们在电气上不同
- 每层的图形由顶视图表示

CMOS工艺层

CMOS工艺层

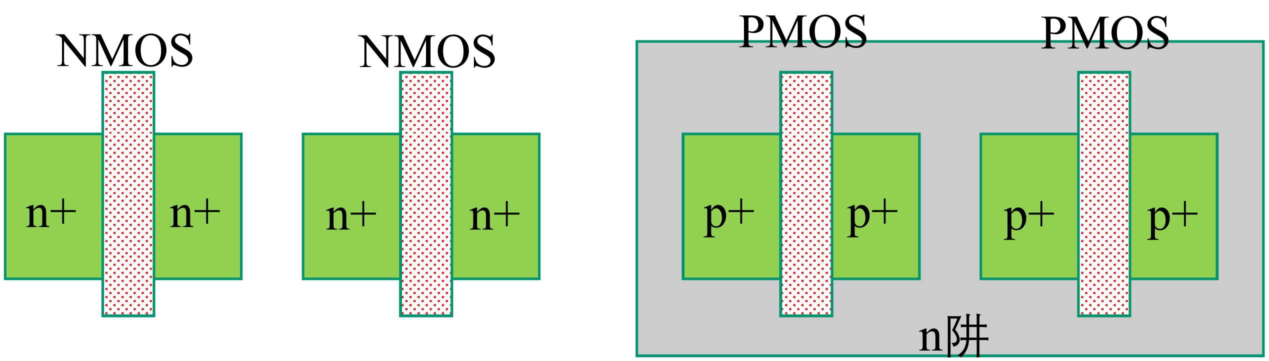


n阱CMOS工艺的截面图

- **p**衬底
- **n**阱
- **n+**（**NMOS**漏/源）
- **p+**（**PMOS**漏/源）
- 栅氧化层
- 栅（多晶硅）

CMOS工艺层

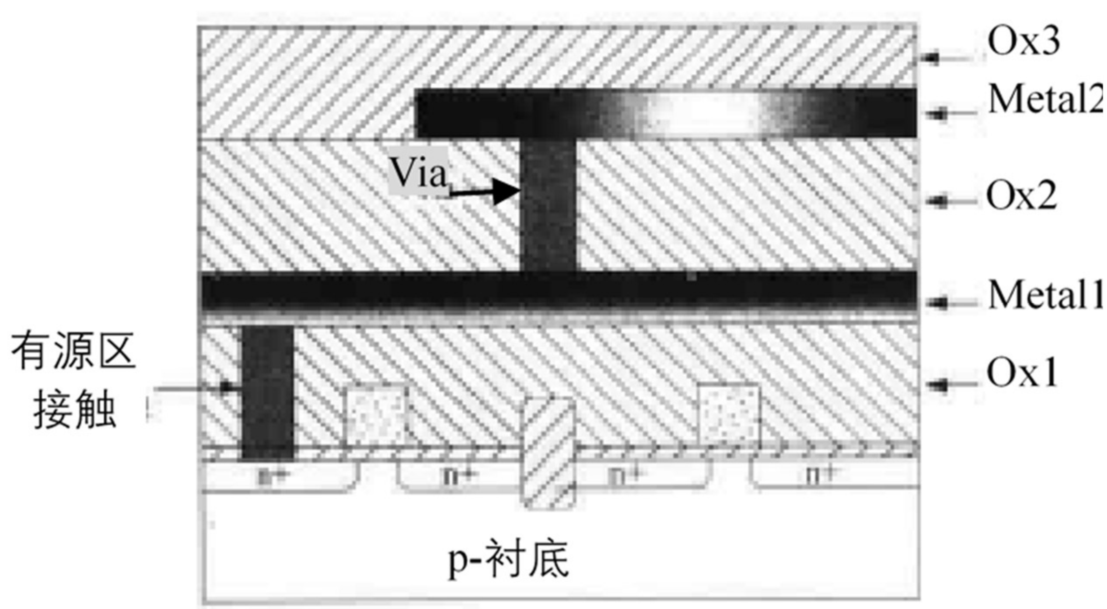
CMOS工艺层



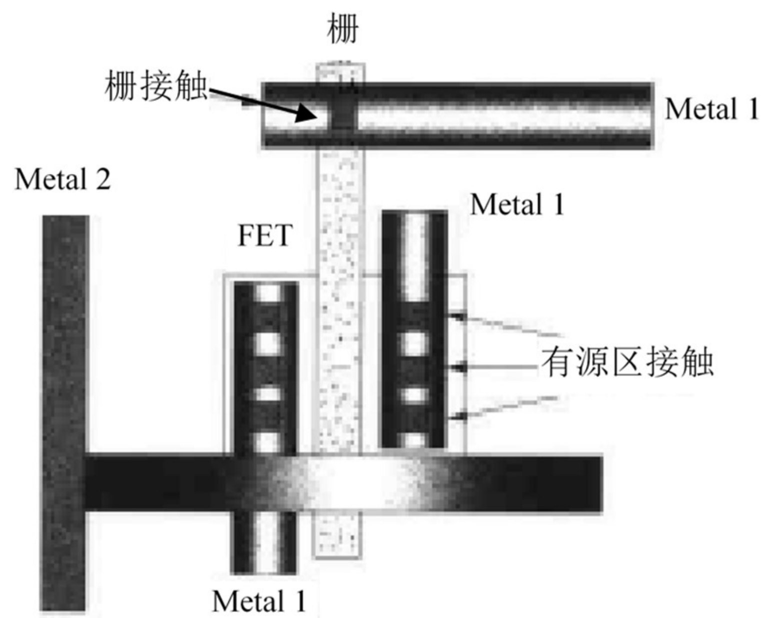
FET图形顶视图

CMOS工艺层

CMOS工艺层



金属互连层



互连线的版图例子

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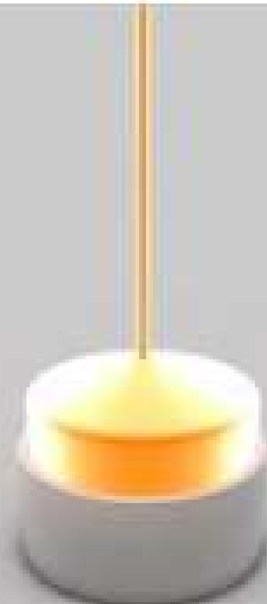
Sand-to-Silicon Video

Sand / Ingot

Sand

With about 25% (mass) Silicon is – after Oxygen – the second most frequent chemical element in the earth's crust. Sand – especially Quartz – has high percentages of Silicon in the form of Silicon dioxide (SiO_2) and is the base ingredient for semiconductor manufacturing.



Melted Silicon –

scale: *wafer level (~300mm / 12 inch)*

Silicon is purified in multiple steps to finally reach semiconductor manufacturing quality which is called Electronic Grade Silicon. Electronic Grade Silicon may only have one alien atom every one billion Silicon atoms. In this picture you can see how one big crystal is grown from the purified silicon melt. The resulting mono crystal is called Ingot.

Mono-crystal Silicon Ingot –

scale: *wafer level (~300mm / 12 inch)*

An ingot has been produced from Electronic Grade Silicon. One ingot weights about 100 kilograms (=220 pounds) and has a Silicon purity of 99.9999%.

Ingot / Wafer

***Ingot Slicing* –**
scale: wafer level (~300mm / 12 inch)
The Ingot is cut into individual silicon discs called wafers.

Wafer –
scale: wafer level (~300mm / *12 inch*)
The wafers are polished until they have flawless, mirror-smooth surfaces. Intel buys those manufacturing ready wafers from third party companies. Intel's highly advanced 45nm High-K/Metal Gate process uses wafers with a diameter of 300 millimeter (~12 inches). When Intel first began making chips, the company printed circuits on 2-inch (50mm) wafers. Now the company uses 300mm wafers, resulting in decreased costs per chip.

Photo Lithography

Applying Photo Resist –
scale: wafer level (~300mm / 12 inch)
The liquid (blue here) that's poured onto the wafer while it spins is a photo resist finish similar as the one known from film photography. The wafer spins during this step to allow very thin and even application of this photo resist layer.

Exposure –
scale: wafer level (~300mm / 12 inch)
The photo resist finish is exposed to ultra violet (UV) light. The chemical reaction triggered by that process step is similar to what happens to film material in a film camera the moment you press the shutter button. The photo resist finish that's exposed to UV light will become soluble. The exposure is done using masks that act like stencils in this process step. When used with UV light, masks create the various circuit patterns on each layer of the microprocessor. A lens (middle) reduces the mask's image. So what gets printed on the wafer is typically four times smaller linearly than the mask's pattern.

Exposure –
scale: transistor level (~50-200nm)
Although usually hundreds of microprocessors are built on a single wafer, this picture story will only focus on a small piece of a microprocessor from now on – on a transistor or parts thereof. A transistor acts as a switch, controlling the flow of electrical current in a computer chip. Intel researchers have developed transistors so small that about 30 million of them could fit on the head of a pin.

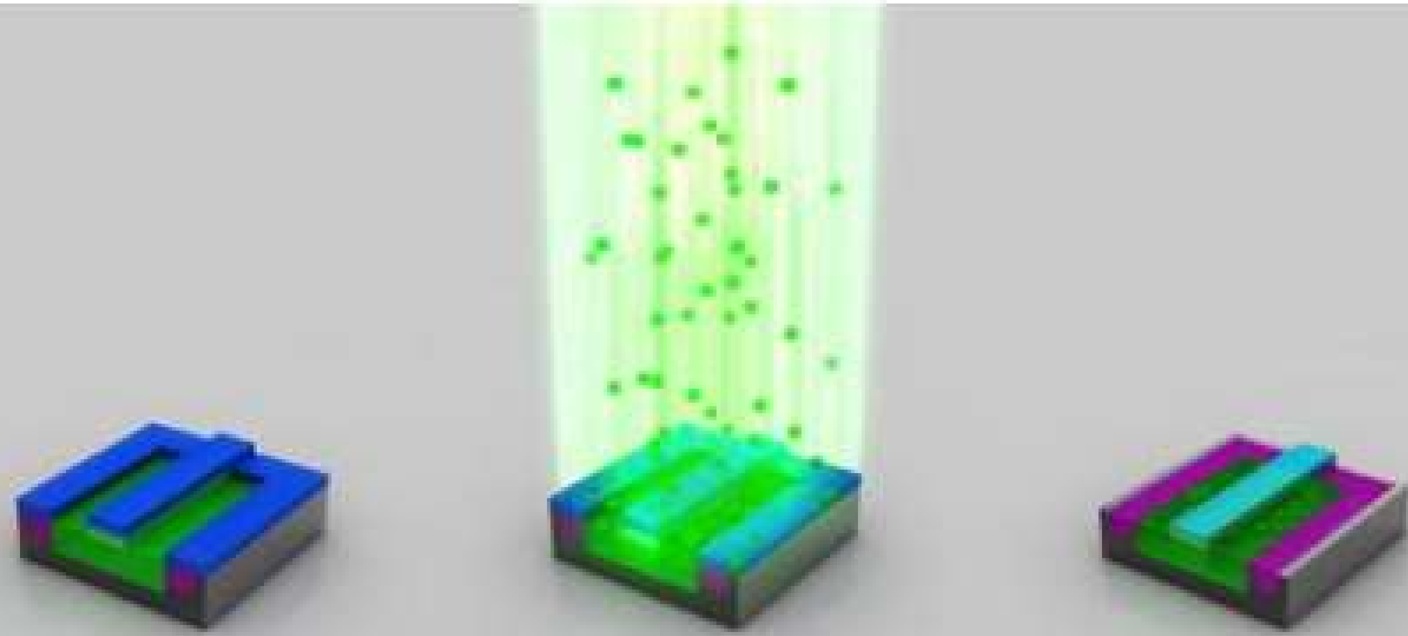
Etching

Washing off of Photo Resist **-**
scale: transistor level (~50-200nm)
The gooey photo resist is completely dissolved by a solvent. This reveals a pattern of photo resist made by the mask.

Etching **-**
scale: transistor level (~50-200nm)
The photo resist is protecting material that should not be etched away. Revealed material will be etched away with chemicals.

Removing Photo Resist **-**
scale: transistor level (~50-200nm)
After the etching the photo resist is removed and the desired shape becomes visible.

Ion Implantation



Applying Photo Resist –
scale: transistor level (~50-200nm)
There's photo resist (blue color) applied, exposed and exposed photo resist is being washed off before the next step. The photo resist will protect material that should not get ions implanted.

Ion Implantation –
scale: transistor level (~50-200nm)
Through a process called ion implantation (one form of a process called doping), the exposed areas of the silicon wafer are bombarded with various chemical impurities called Ions. Ions are implanted in the silicon wafer to alter the way silicon in these areas conducts electricity. Ions are shot onto the surface of the wafer at very high speed. An electrical field accelerates the ions to a speed of over 300,000 km/h (~185,000 mph)

Removing Photo Resist –
scale: transistor level (~50-200nm)
After the ion implantation the photo resist will be removed and the material that should have been doped (green) has alien atoms implanted now (notice slight variations in color)

Metal Deposition

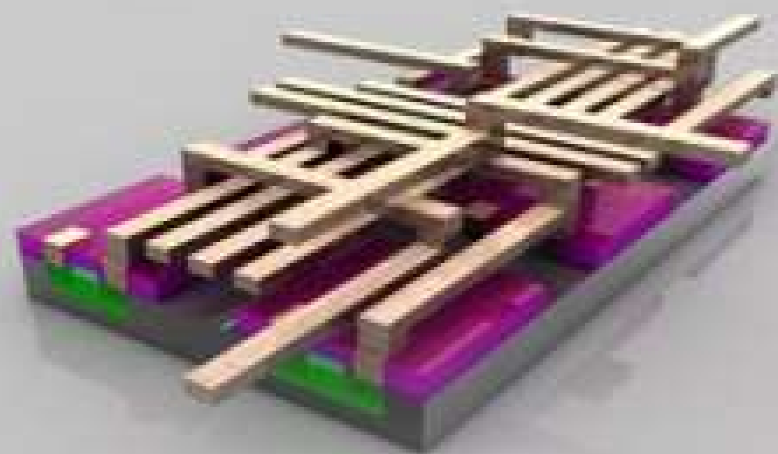
Ready Transistor –
scale: transistor level (~50-200nm)
This transistor is close to being finished. Three holes have been etched into the insulation layer (magenta color) above the transistor. These three holes will be filled with copper which will make up the connections to other transistors.

Electroplating –
scale: transistor level (~50-200nm)
The wafers are put into a copper sulphate solution as this stage. The copper ions are deposited onto the transistor thru a process called electroplating. The copper ions travel from the positive terminal (anode) to the negative terminal (cathode) which is represented by the wafer.

After Electroplating –
scale: transistor level (~50-200nm)
On the wafer surface the copper ions settle as a thin layer of copper.

Metal Layers

Polishing –
scale: transistor level (~50-200nm)
The excess material is polished off.



Metal Layers – *scale: transistor level (six transistors combined ~500nm)*
Multiple metal layers are created to interconnect (think: wires) in between the various transistors. How these connections have to be "wired" is determined by the architecture and design teams that develop the functionality of the respective processor (e.g. Intel® Core™ i7 Processor). While computer chips look extremely flat, they may actually have over 20 layers to form complex circuitry. If you look at a magnified view of a chip, you will see an intricate network of circuit lines and transistors that look like a futuristic, multi-layered highway system.

Wafer Sort Test / Slicing

Wafer Sort Test –
scale: die level (~10mm / ~0.5 inch)
This fraction of a ready wafer is being put to a first functionality test. In this stage test patterns are fed into every single chip and the response from the chip monitored and compared to "the right answer".

Wafer Slicing –
scale: wafer level (~300mm / 12 inch)
The wafer is cut into pieces (called dies).

Discarding faulty Dies –
scale: wafer level (~300mm / 12 inch)
The dies that responded with the right answer to the test pattern will be put forward for the next step (packaging).

Packaging

Individual Die -
scale: die level (~10mm / ~0.5 inch)
This is an individual die which has been cut out in the previous step (slicing). The die shown here is a die of an Intel® Core™ i7 Processor .

Packaging -
scale: package level (~20mm / ~1 inch)
The substrate, the die and the heatspreader are put together to form a completed processor. The green substrate builds the electrical and mechanical interface for the processor to interact with the rest of the PC system. The silver heatspreader is a thermal interface where a cooling solution will be put on to. This will keep the processor cool during operation.

Processor -
scale: package level (~20mm / ~1 inch)
Completed processor (Intel® Core™ i7 Processor in this case). A microprocessor is the most complex manufactured product on earth. In fact, it takes hundreds of steps – only the most important ones have been visualized in this picture story - in the world's cleanest environment (a microprocessor fab) to make microprocessors.

Class Testing / Completed Processor

Class Testing -
scale: package level (~20mm / ~1 inch)
During this final test the processors will be tested for their key characteristics (among the tested characteristics are power dissipation and maximum frequency).

Binning -
scale: package level (~20mm / ~1 inch)
Based on the test result of class testing processors with the same capabilities are put into the same transporting trays.

Retail Package -
scale: package level (~20mm / ~1 inch)
The readily manufactured and tested processors (again Intel® Core™ i7 Processor is shown here) either go to system manufacturers in trays or into retail stores in a box such as that shown here.

制造

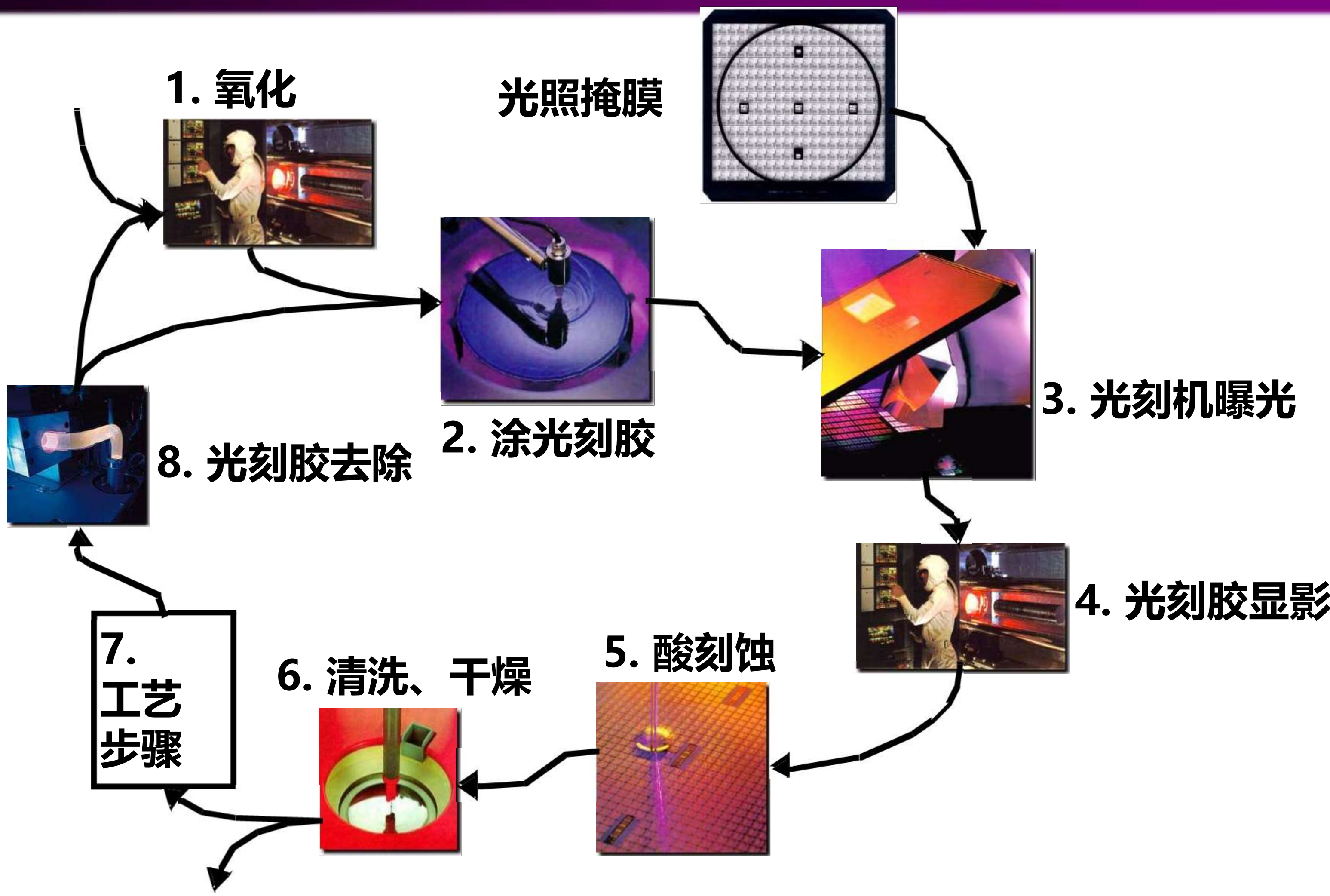
- 制造芯片的巨型工厂称为**fabs**
- 有很大的超净间（类似足球场大小）

Courtesy of International Business Machines Corporation. Unauthorized use not permitted.

主要的工艺步骤

- 扩散和离子注入
  - 源、漏、阱及衬底接触形成
  - 多晶掺杂及器件阈值调整
- 材料的生长与淀积
  - 二氧化硅（**SiO2**）**—** 热氧化或淀积
  - 氮化硅（$\mathbf{Si_3N_4}$）— 化学气相淀积（**CVD**）
  - 多晶—化学淀积
  - 铝互连—溅射
- 刻蚀
  - 湿法刻蚀
  - 干法刻蚀或等离子刻蚀
- 平面化—化学机械抛光（**CMP**）

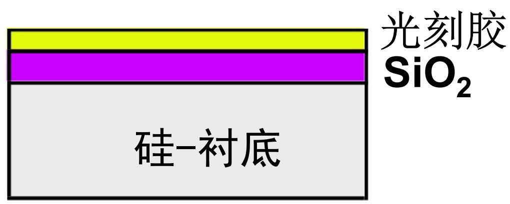
光刻过程



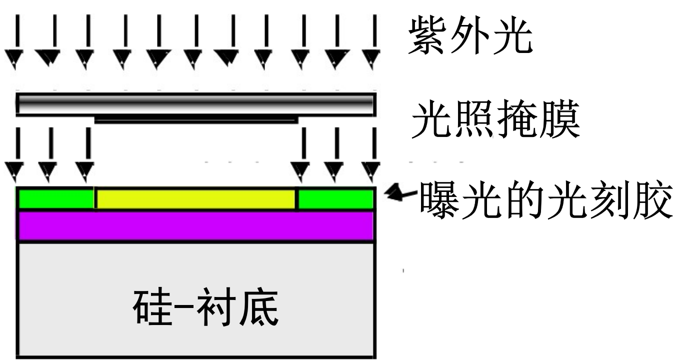
一次光刻过程的典型操作步骤

光刻过程举例：形成 SiO_2图形

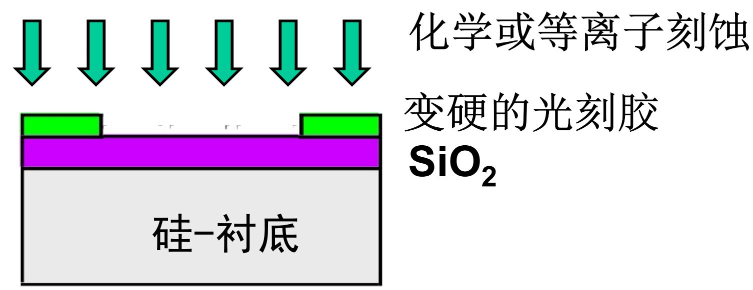
(a) 硅基础材料



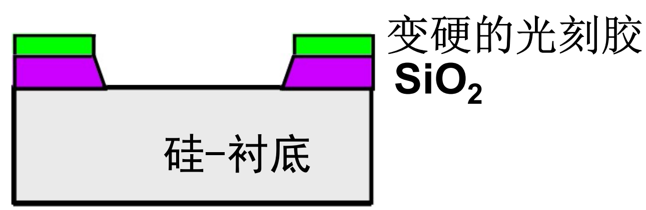
(b) 氧化及淀积负光刻胶后



(c) 光刻机曝光



(d) 显影和刻蚀光刻胶后刻蚀SiO_2



(e) SiO_2刻蚀后

(f) 去除光刻胶后的最终结果

CMOS制造步骤

制造**CMOS**电路工艺顺序（概略）

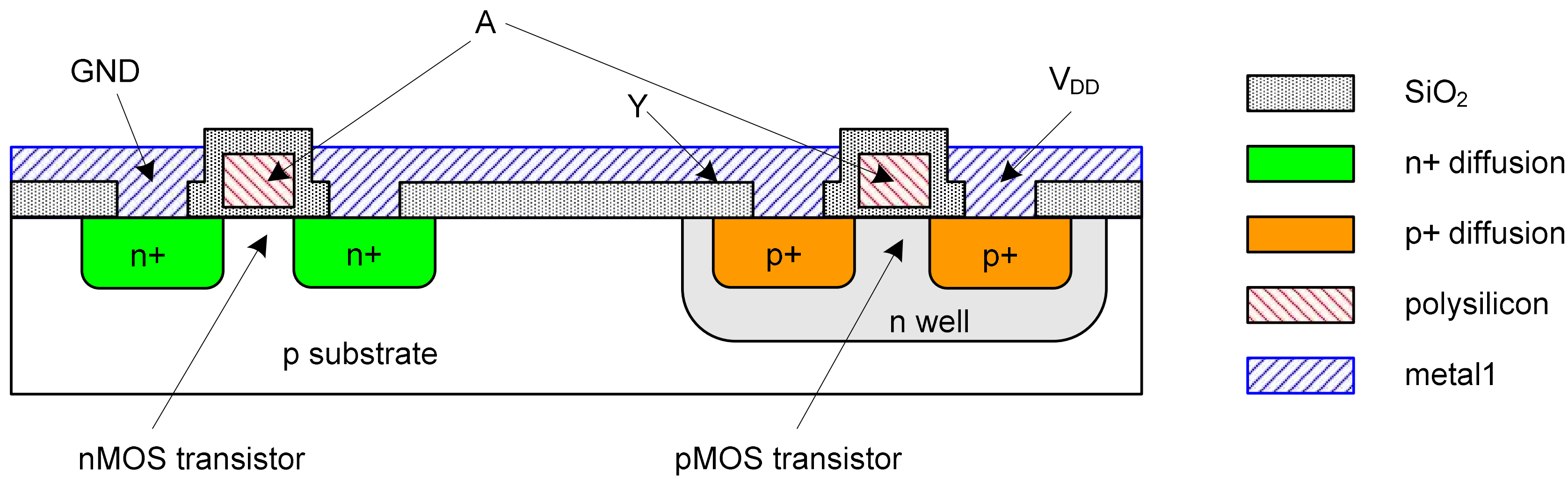
定义有源区刻蚀及在绝缘沟槽中填充氧化物
↓
阱区离子注入
↓
淀积及形成多晶硅图形
↓
源区、漏区和衬底接触的离子注入
↓
形成接触和通孔窗口
淀积及形成金属层图形

CMOS 制造

- **CMOS** 晶体管基于硅圆片（**wafer**）进行制造
- 光刻过程类似于印刷机印刷过程
- 每一步工艺淀积或刻蚀不同的材料
- 便于理解，采用简单制造工艺的顶视图和横切图演示说明

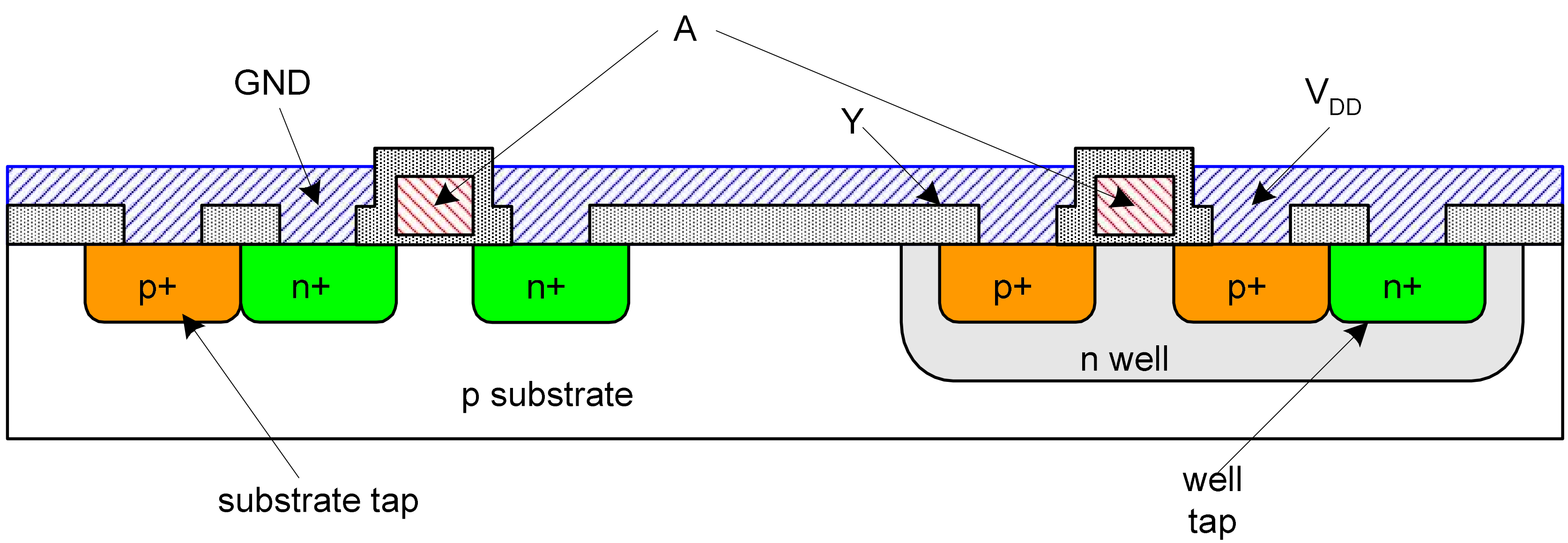
反相器横切图

- 通常用**P**型衬底制造**nMOS**
- 需要用**n-**阱（**n-well**）作为**pMOS**晶体管的衬底



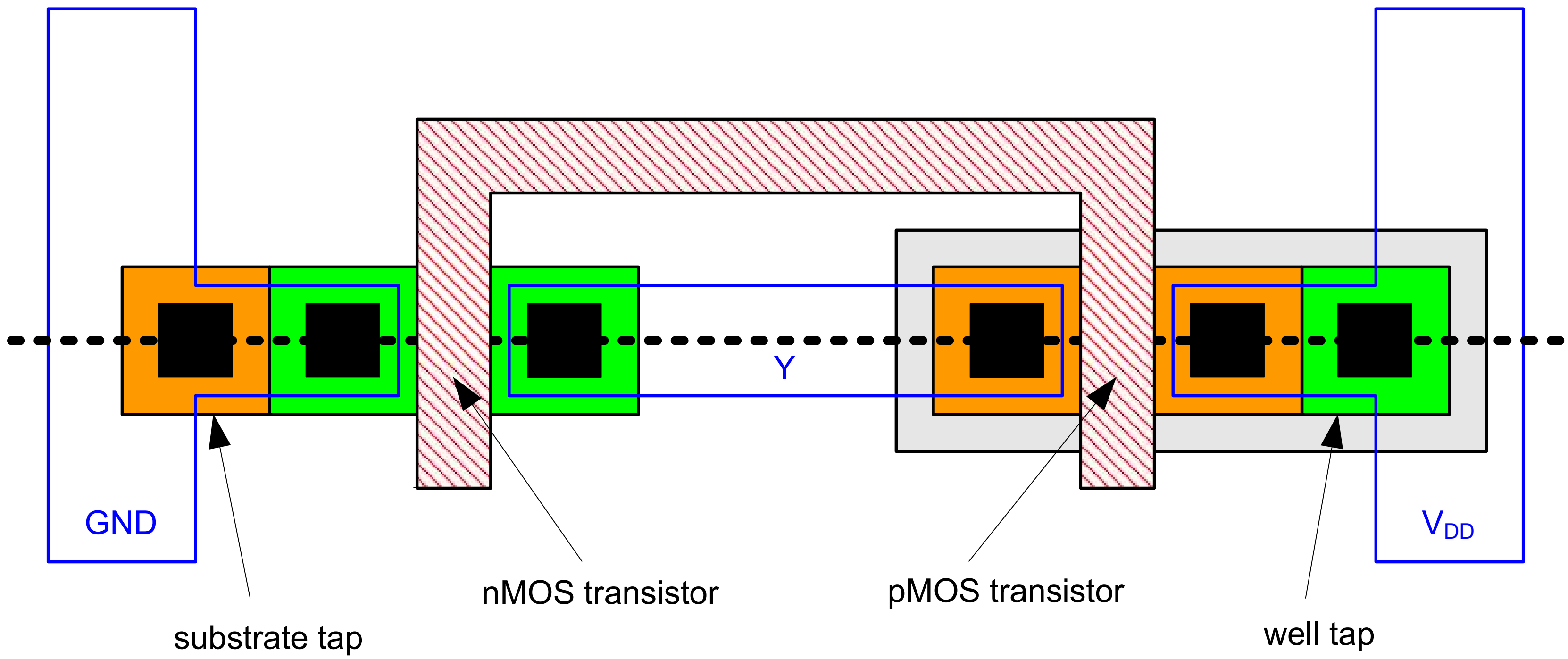
阱和衬底

- 衬底需要连接GND， 而n-well 需要连接 V_{DD}
- 金属与轻掺杂半导体接触形成弱连接，称为肖特基二极管（Shottky Diode）
- 需要用重掺杂区形成阱和衬底与金属接触



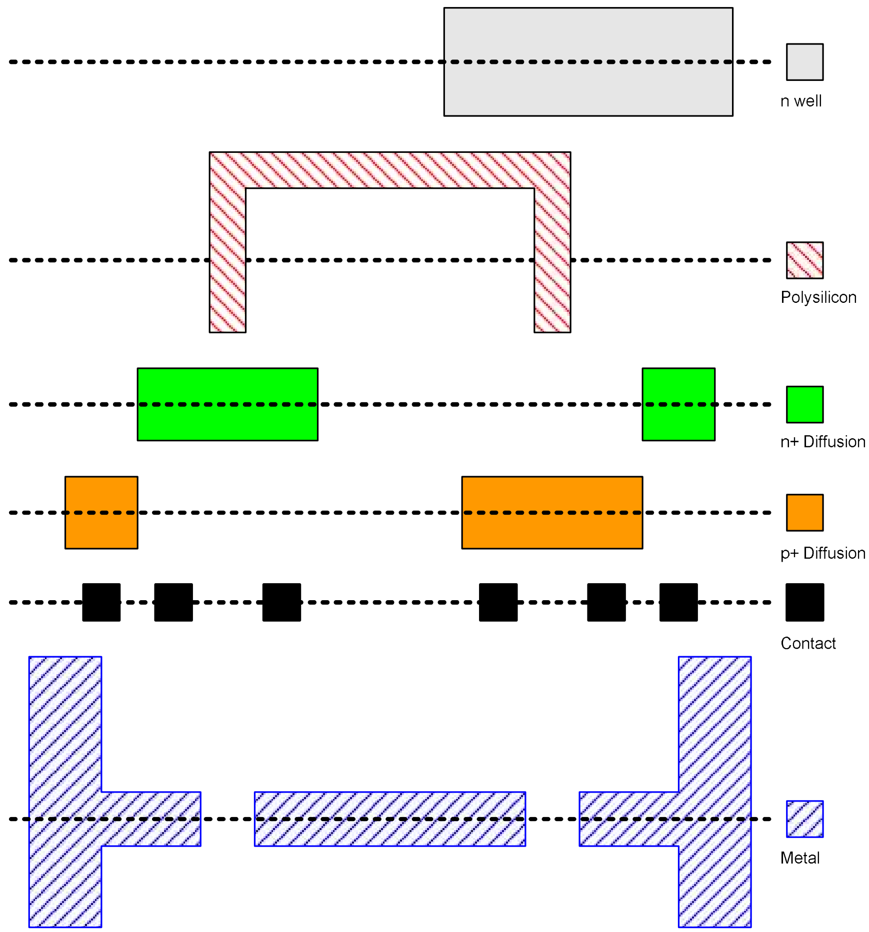
反相器掩膜组（Mask Set）

- 晶体管和连线由掩膜定义
- 沿下图虚线横切



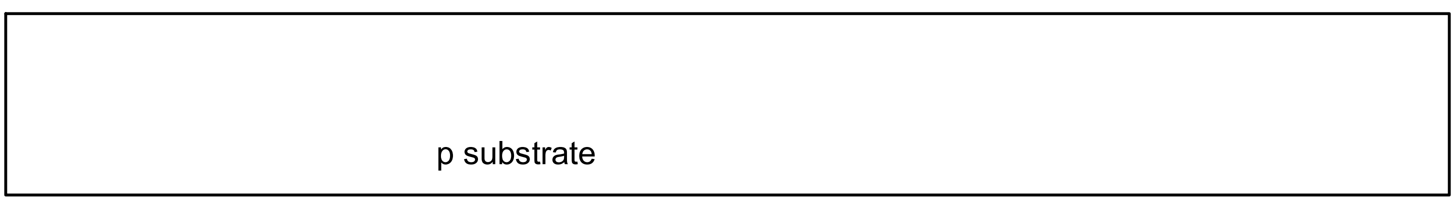
详细掩膜图

- 6层主要掩膜
  - n-well n阱
  - Polysilicon 多晶
  - n+ diffusion
  - p+ diffusion
  - Contact 接触孔
  - Metal 金属



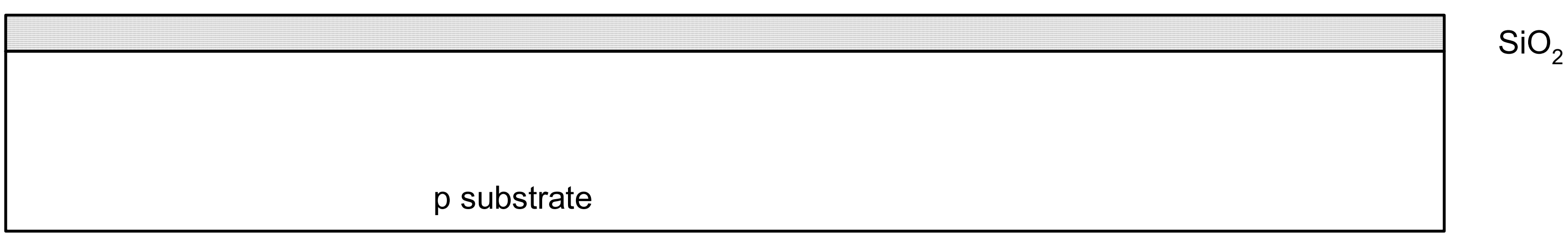
制造步骤

- 从空白的**wafer**开始
- 从底层往上构造反相器
- 第一步形成**n-well**
  - **Wafer**表面覆盖 **(oxide)**保护层
  - 去除需要形成**n-well**处的SiO_2 保护层
  - **N**型参杂物注入或扩散到**wafer**无保护层的区域内
  - 去除**wafer**表面剩余的SiO_2



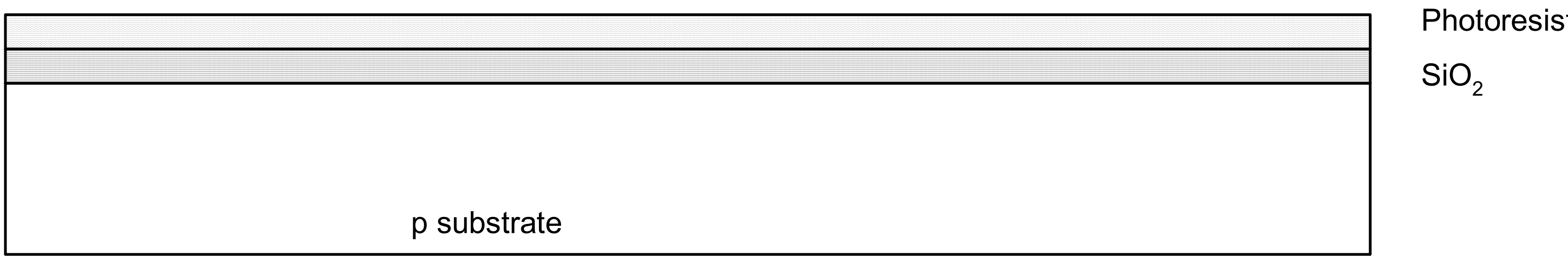
氧化（Oxidation）

- 硅圆片（wafer）生长SiO_2
  - 900 – 1200 °C氧化炉内通入 H_2O 或 O_2



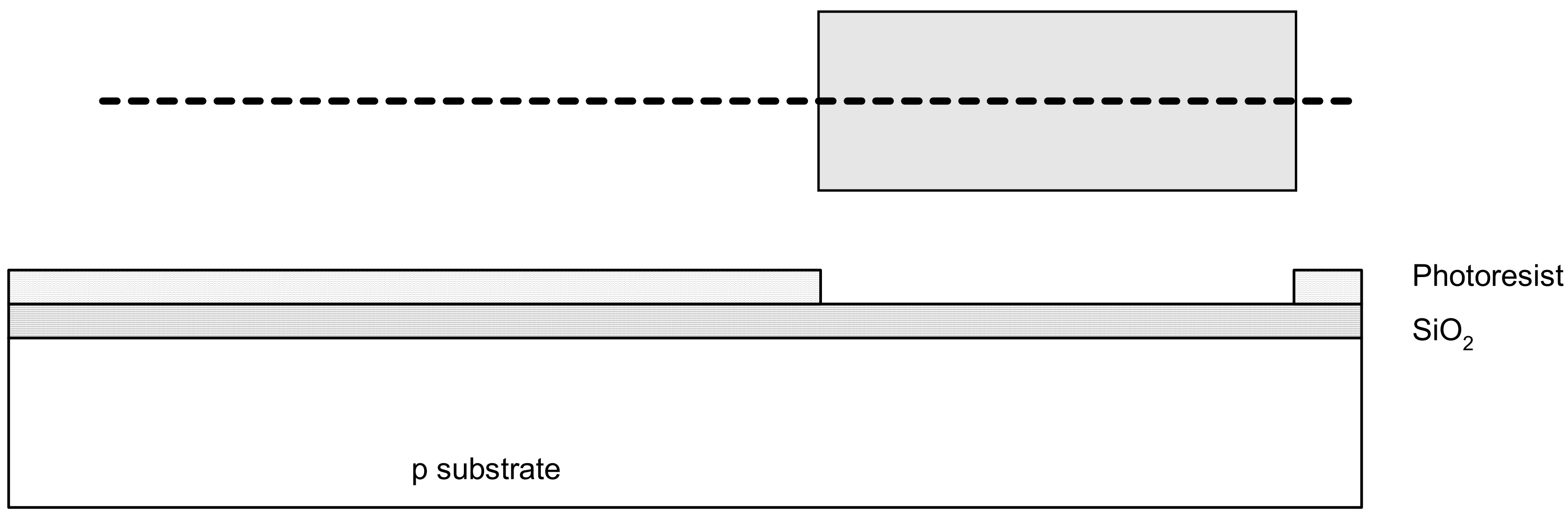
光刻胶（Photoresist）

- 旋转涂光刻胶
  - 光刻胶是一种光敏的有机聚合物
  - 曝光的部分变硬



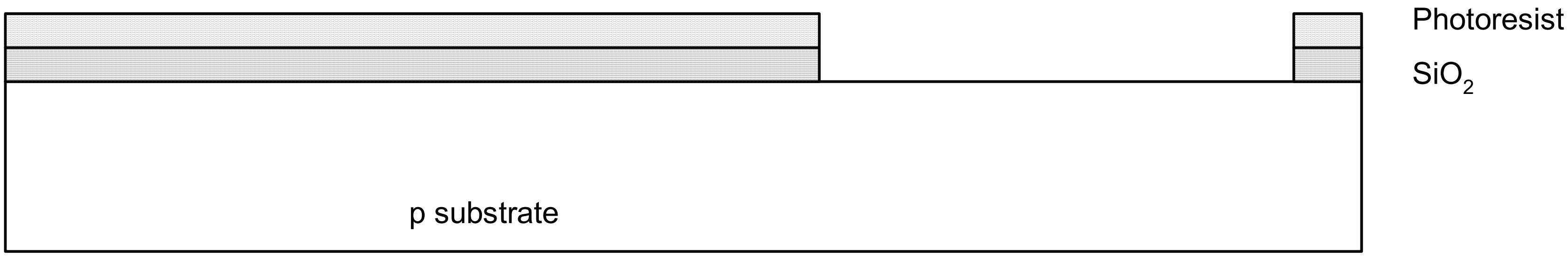
光刻（Lithography）

- 用**n-well**掩膜对光刻胶曝光
- 去除未曝光的光刻胶



刻蚀（Etch）

- 用氢氟酸（HF）刻蚀氧化层
- 没有光刻胶保护的氧化层被刻蚀

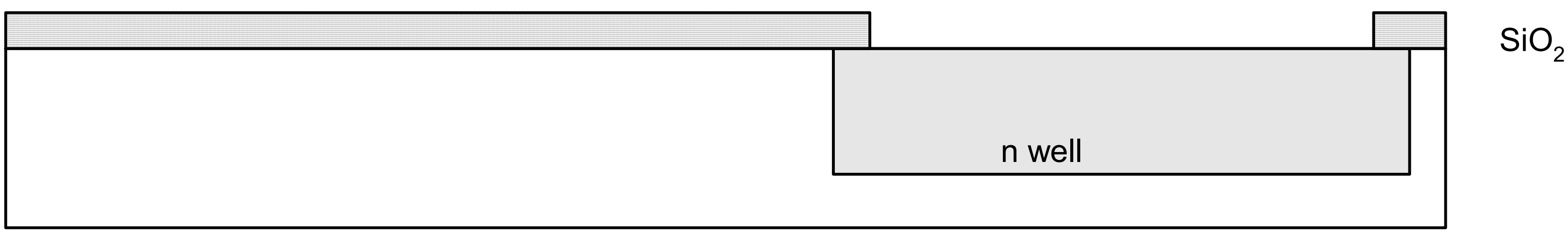


除去光刻胶

- 去除剩余的光刻胶
  - 使用混合酸
- 避免残余光刻胶熔于下一步工艺

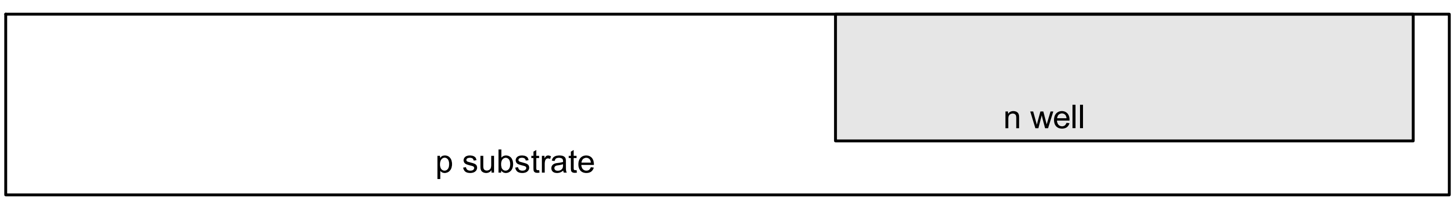
n-well注入

- 用扩散或离子注入形成
- 扩散（**Diffusion**）
  - 圆片放入充砷（**As**）气的炉管内
  - 加热直至砷原子扩散进入暴露的硅内
- 离子注入（**Ion Implanatation**）
  - 用砷离子束轰击圆片**wafer**
  - 离子会被SiO_2阻挡，只进入暴露的硅内



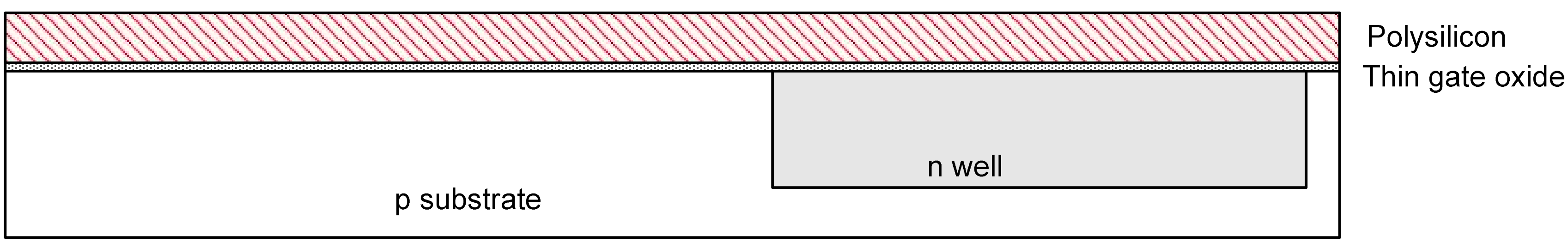
除去氧化层

- 用氢氟酸去除剩余的氧化层
- 回复到带有**n-well**的裸圆片
- 后续的工艺步骤就是一系列类似的过程



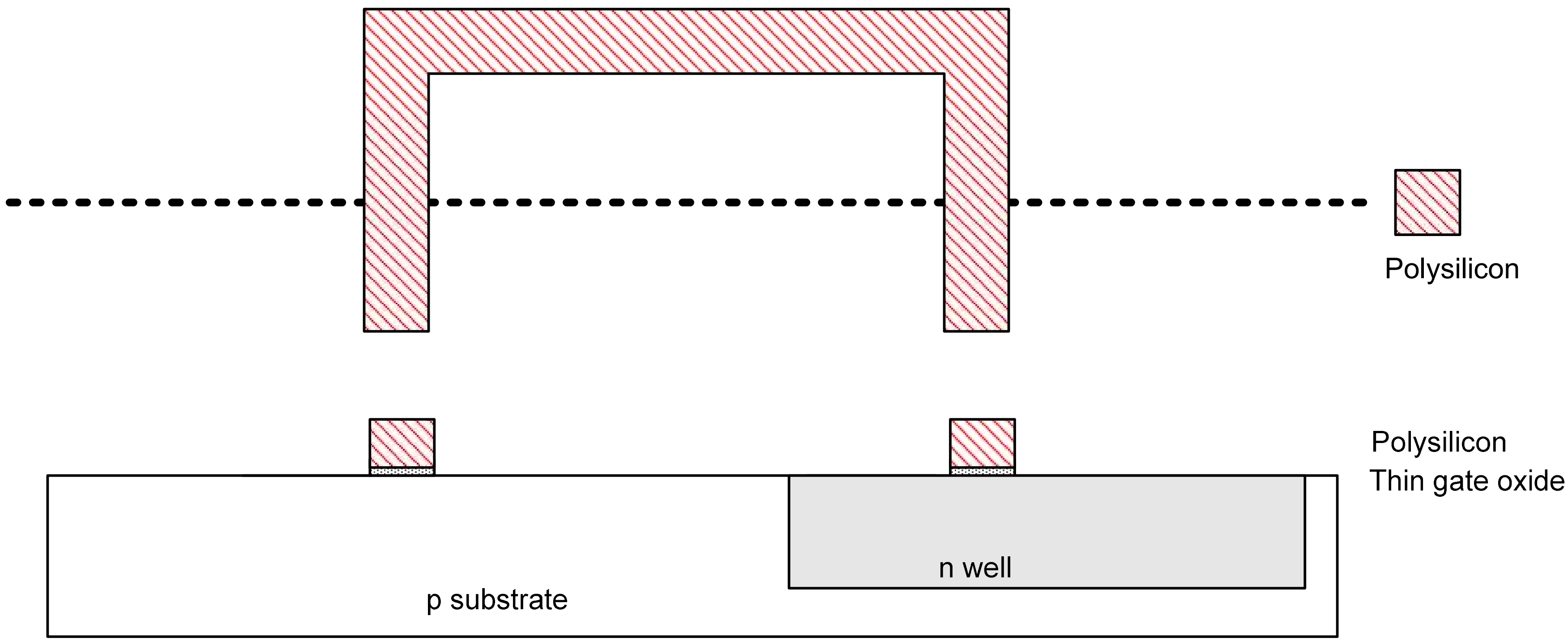
多晶（Polysilicon）

- 生长非常薄的栅氧化层
  - < 20 Å (6-7 atomic layers)
- 多晶硅淀积（CVD，Chemical Vapor Deposition）
  - 圆片放入充满硅烷 (SiH_4) 的炉管
  - 形成很多小的硅晶粒称为多晶硅（polysilicon）
  - 重掺杂多晶硅使之成为良好的导体



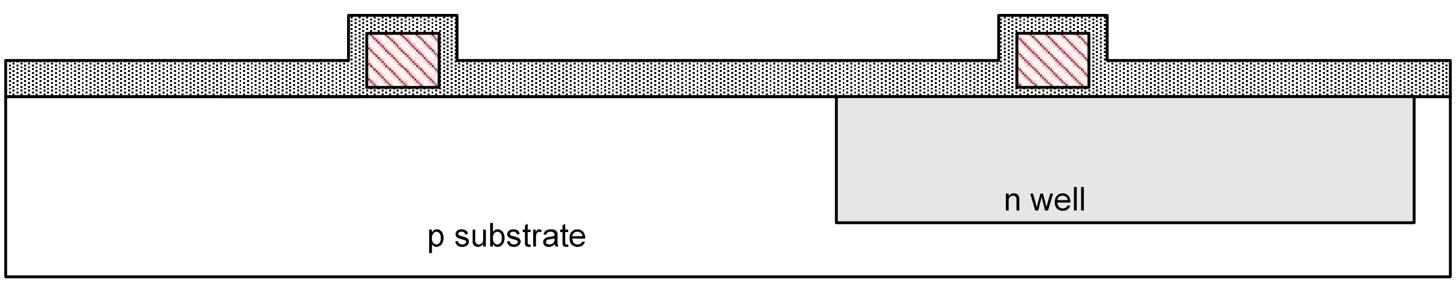
刻蚀多晶

■ 采用类似前面的光刻过程形成多晶硅图案



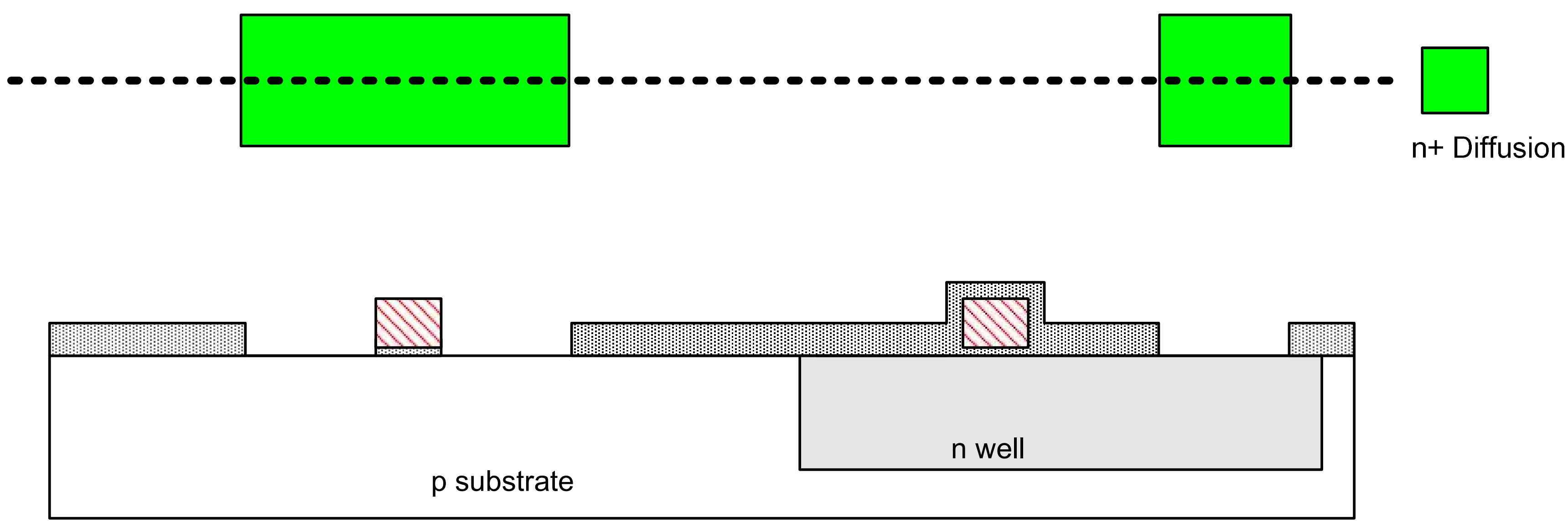
自对准工艺（Self-Aligned Process）

- 采用氧化和掩膜曝光要进行扩散或注入的 **n+** 掺杂区（**N**型扩散区）
- **N**型扩散区形成 **nMOS** 的源、漏和 **n-well** 接触



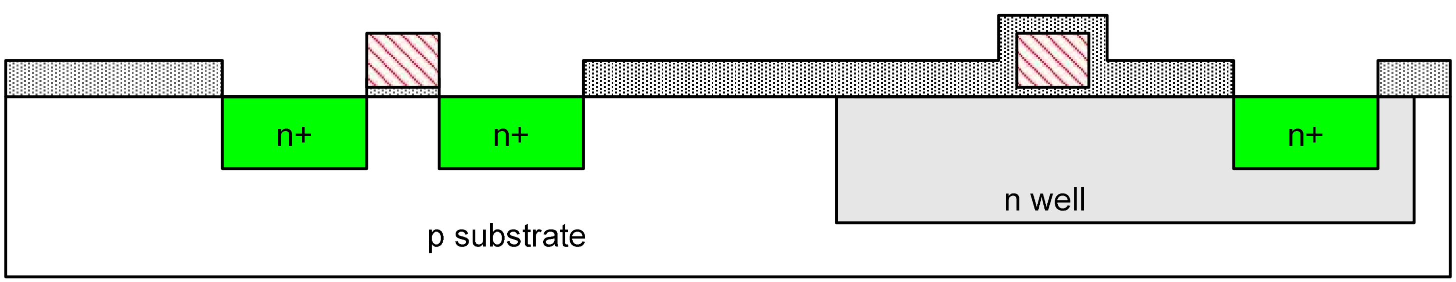
N型扩散区

- 刻氧化层并形成**N**型扩散区
- *自对准工艺*的栅可以阻挡对沟道区的扩散
- 多晶比金属更适合自对准工艺，因为在后续的处理中多晶硅不易熔化



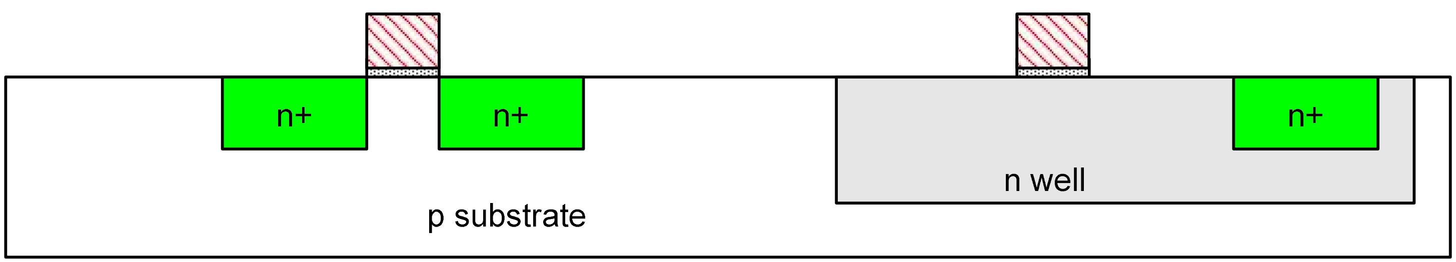
N型扩散区

- 历史上采用扩散进行掺杂
- 现在通常采用离子注入掺杂
- 但这些区域仍称之为扩散区



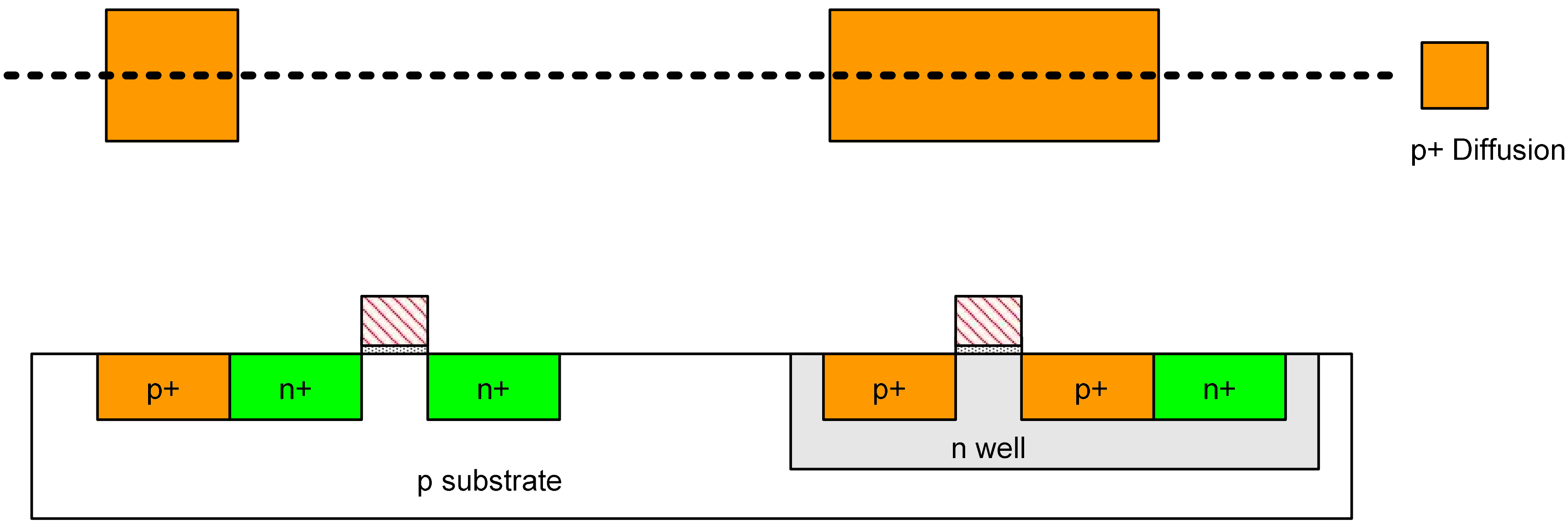
N型扩散区

- 去除氧化层



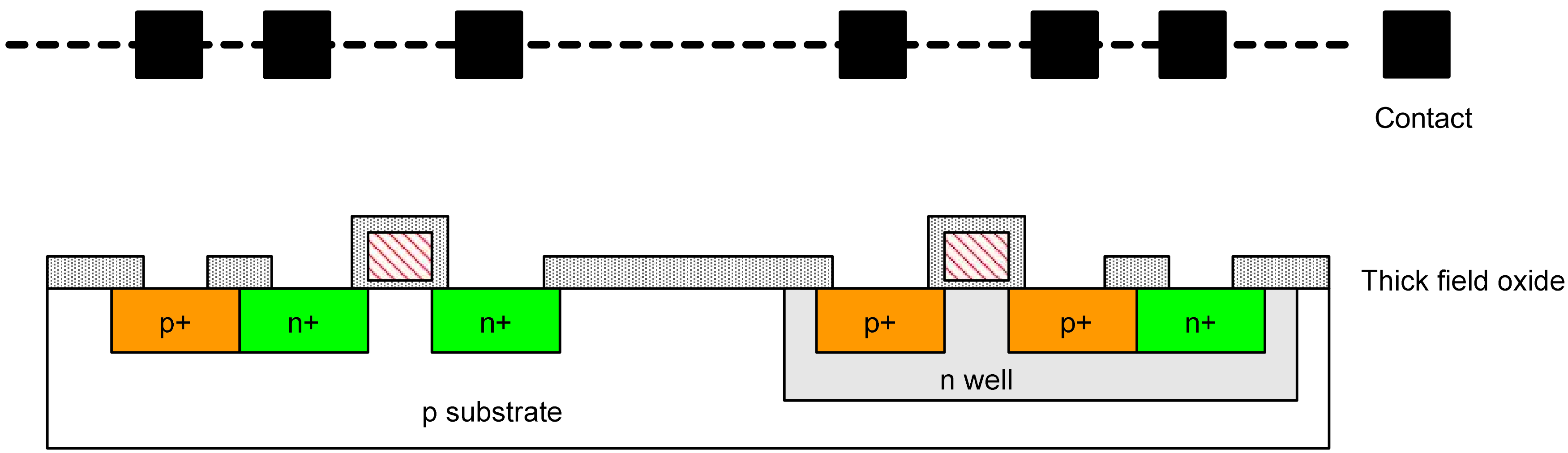
P型扩散区

- 采用类似的一套步骤形成 P型扩散区作为pMOS的源、漏和衬底接触



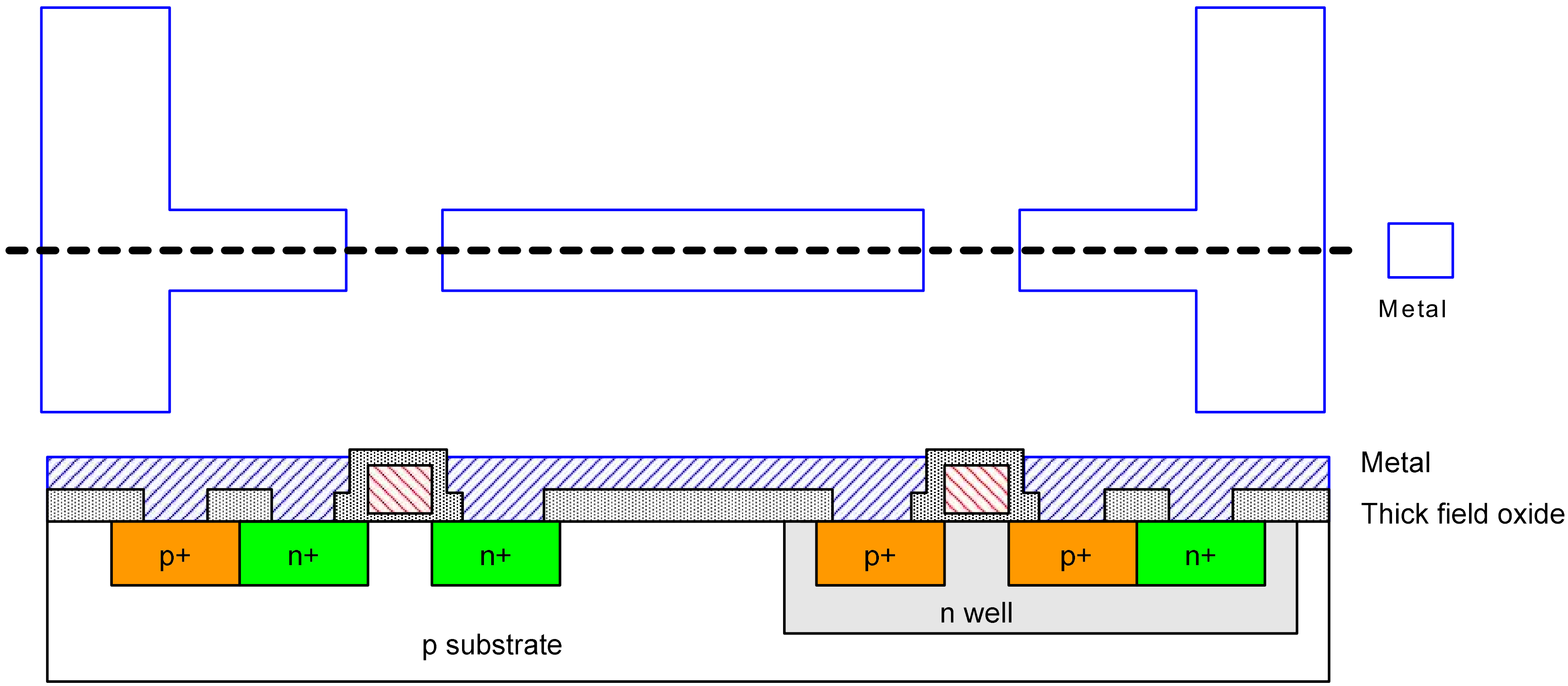
接触孔（Contacts）

- 需要把器件按要求连接起来
- 整个芯片上盖上厚的场氧化层
- 刻蚀去除接触孔处的氧化层

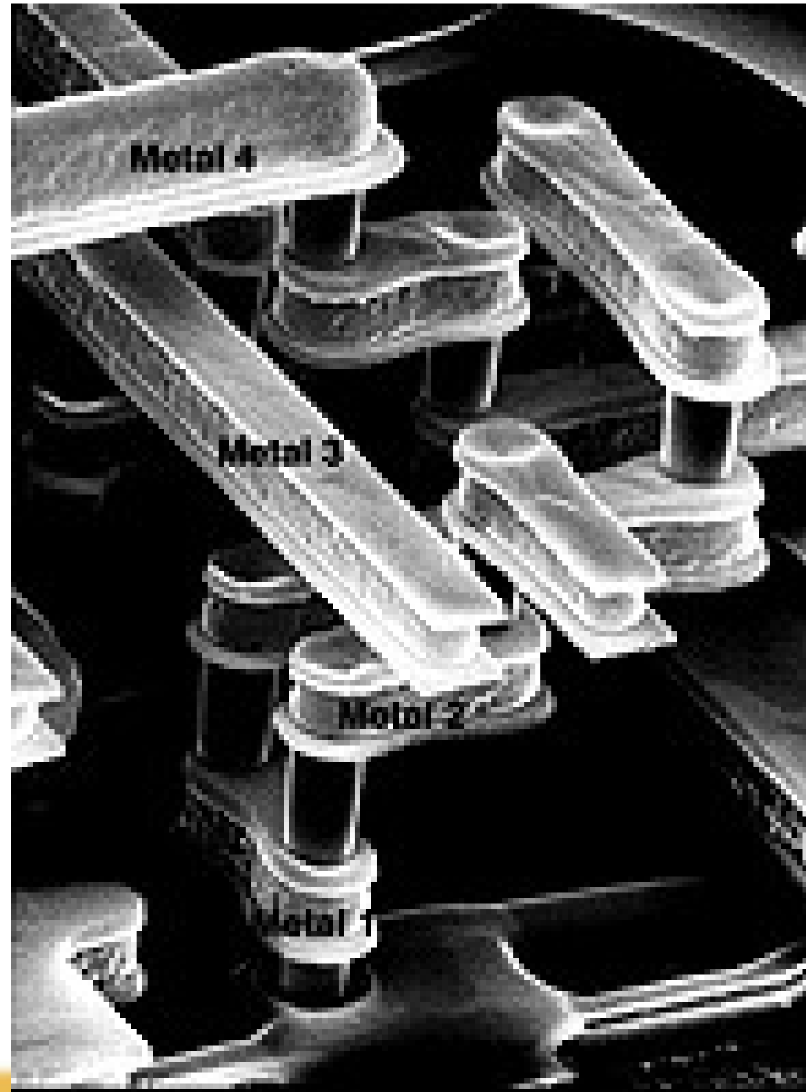


金属化（Metalization）

- 在整个wafer上溅射金属铝Al
- 刻蚀去除多余的金属，保留连线金属



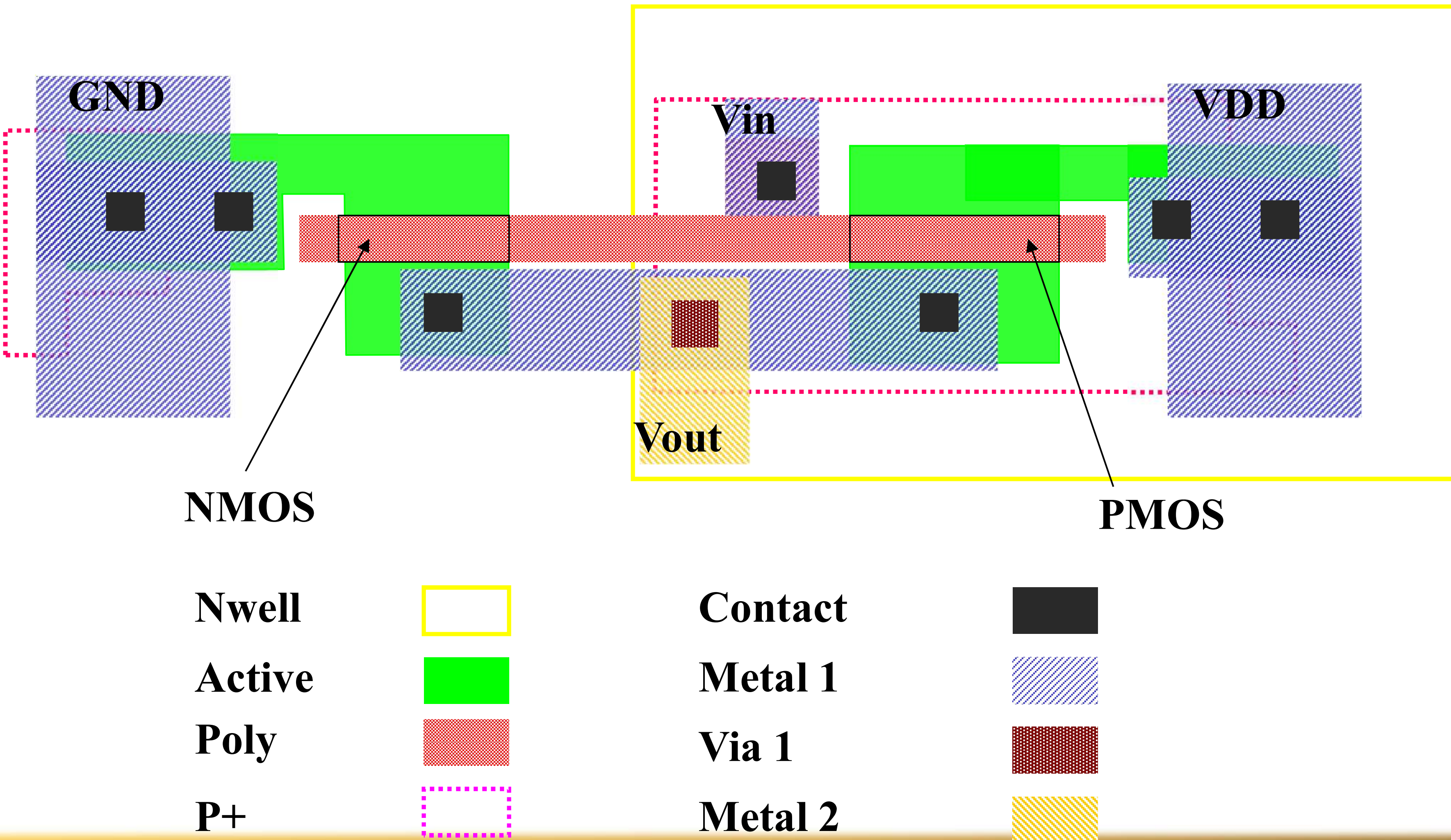
先进金属工艺（Advanced Metallization）



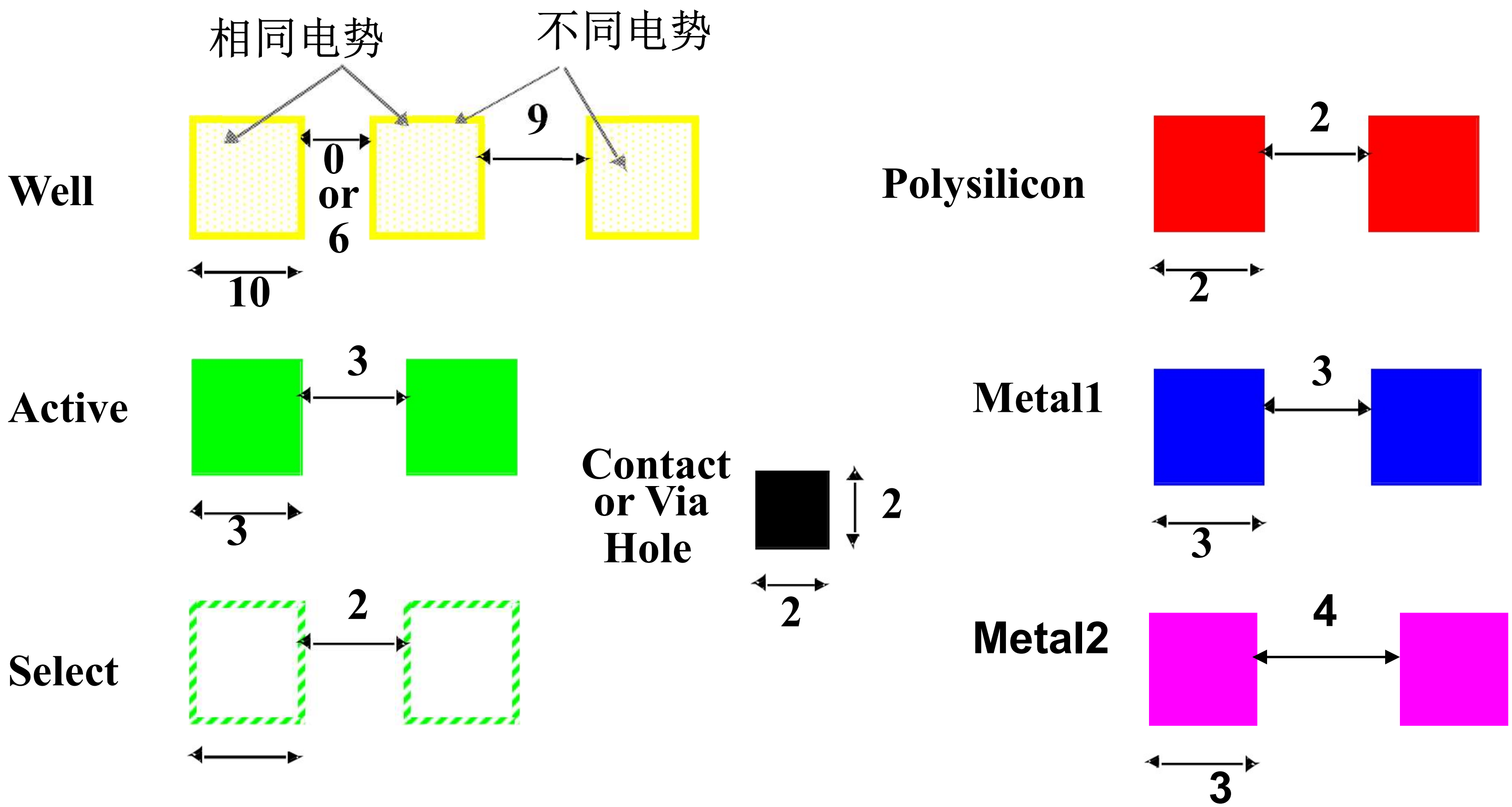
二、设计规则（Design Rules）

- 设计工程师与工艺工程师之间的接口
- 构造工艺掩膜的指导原则
- 单位尺度：最小线宽
  - 等比例缩小的设计规则：**lambda (λ)**参数
  - 绝对尺寸（微米规则）

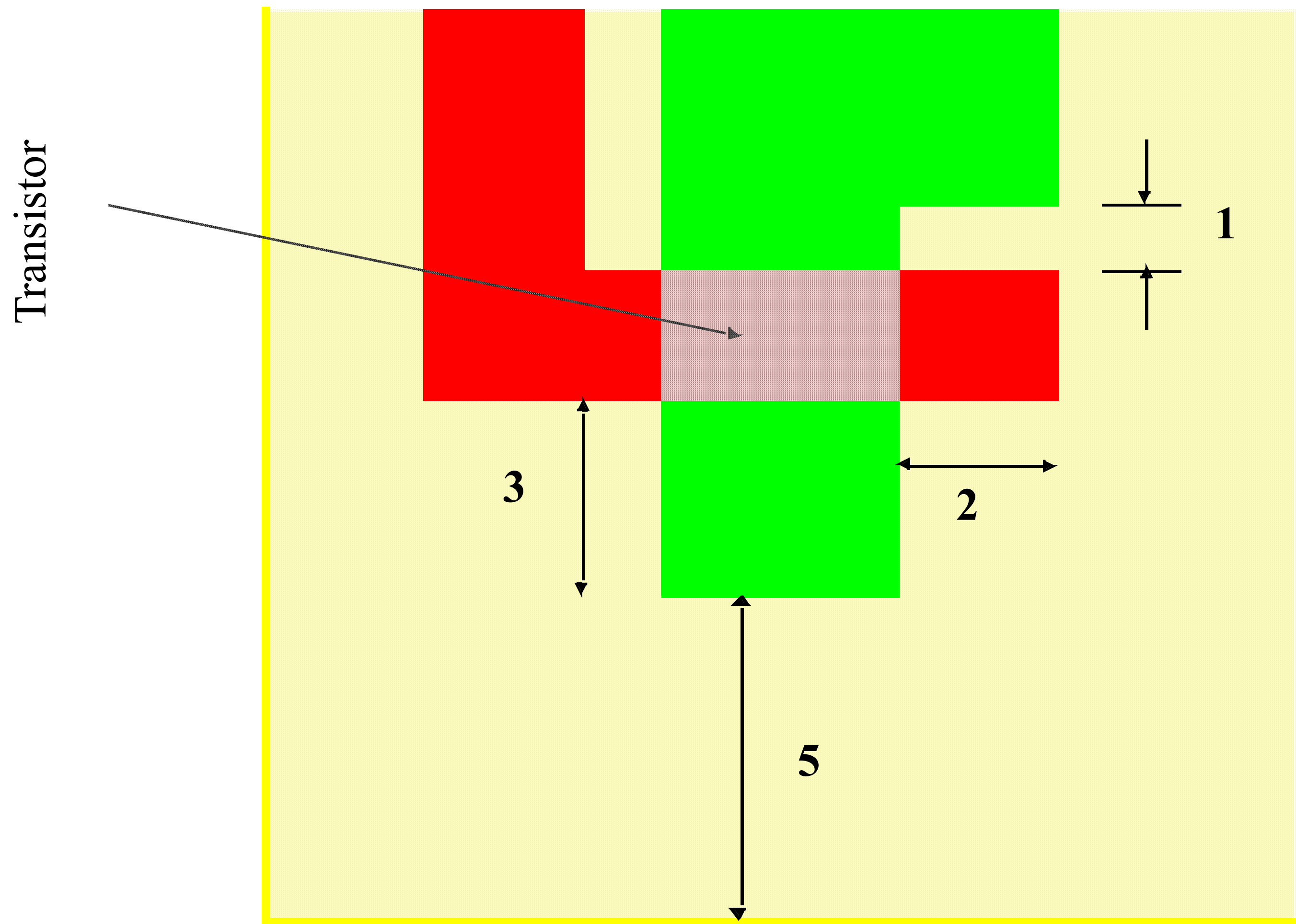
CMOS反相器版图


GND
Vin
VDD
Vout
NMOS
PMOS
Nwell
Active
Poly
P+
Contact
Metal 1
Via 1
Metal 2

层内（Intra-Layer）设计规则



层间设计规则 - 晶体管版图

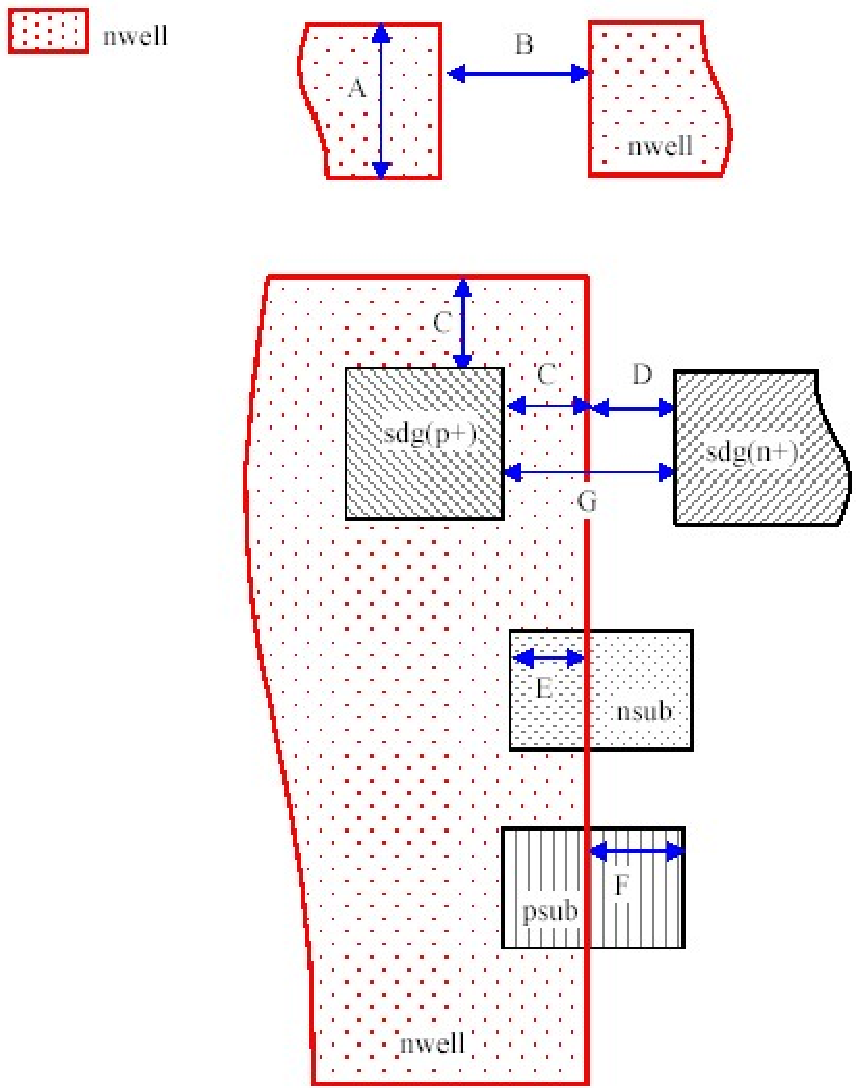


设计规则举例：0.25um CMOS设计规则

层定义

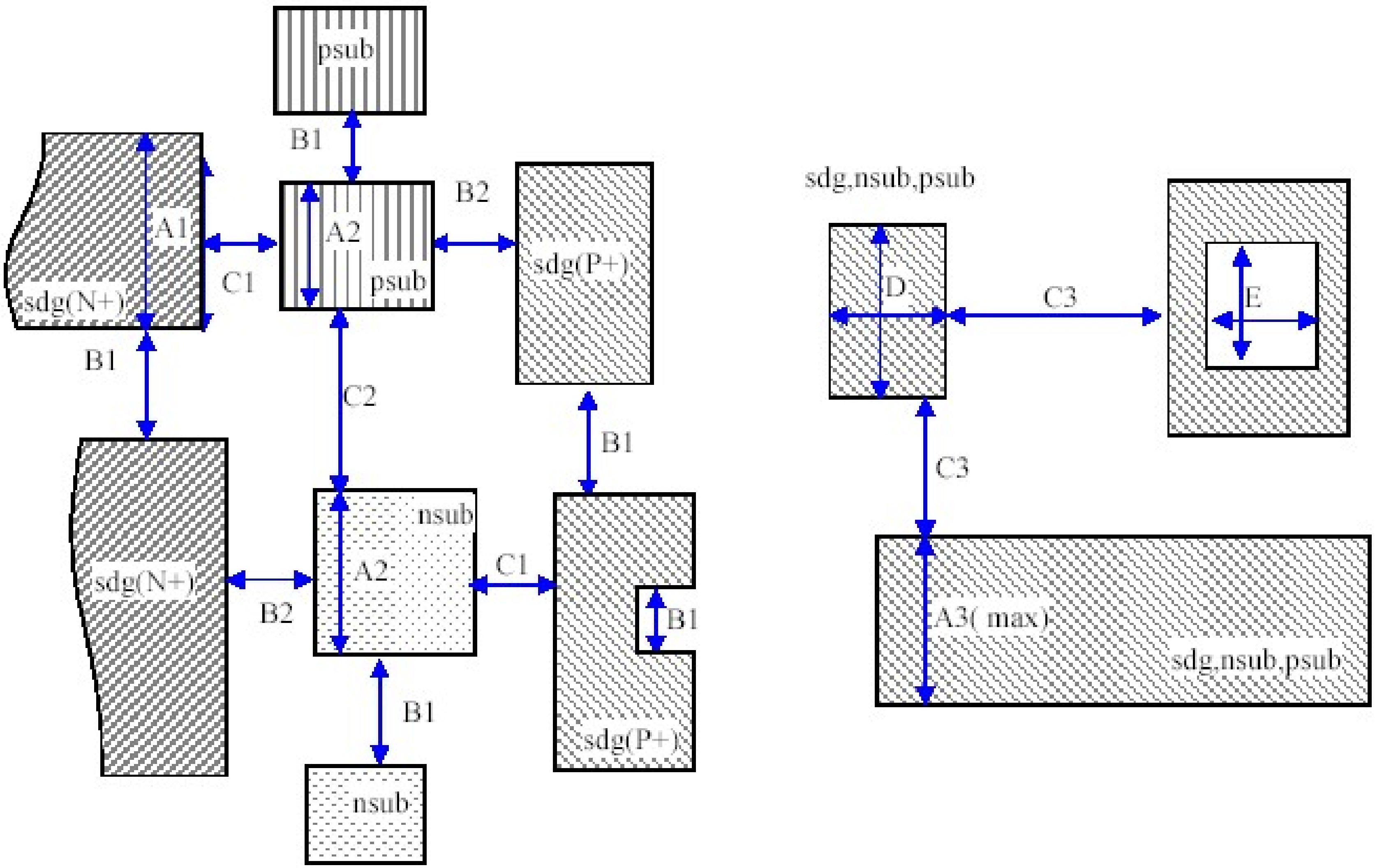
序号	名称	说明
1	nwell	N-Well
2	sdg	Diffusion (Source/Drain/Gate) for Transistor
3	Poly	Poly-Si for gate electrode and interconnection line
4	Contact	Contact between Si and 1st metal
5	Metal1	1st metal layer
6	Via1	via contact between 1st and 2nd metal
7	Metal2	2nd metal layer
8	Via2	via contact between 2nd and 3rd metal
9	Metal3	3rd metal layer
10	Pad	Pad opening

Nwell



序号	描述	限制
L1-A	nwell width	1.76
L1-B	nwell spacing	1.76
L1-C	nwell ~ sdg(P+) margin	0.64
L1-D	nwell ~ sdg(N+) margin	0.64
L1-E	nwell ~ nsub margin	0.40
L1-F	psub ~ nwell margin	0.40
L1-G	sdg(P+) ~ sdg(N+) spacing	1.28

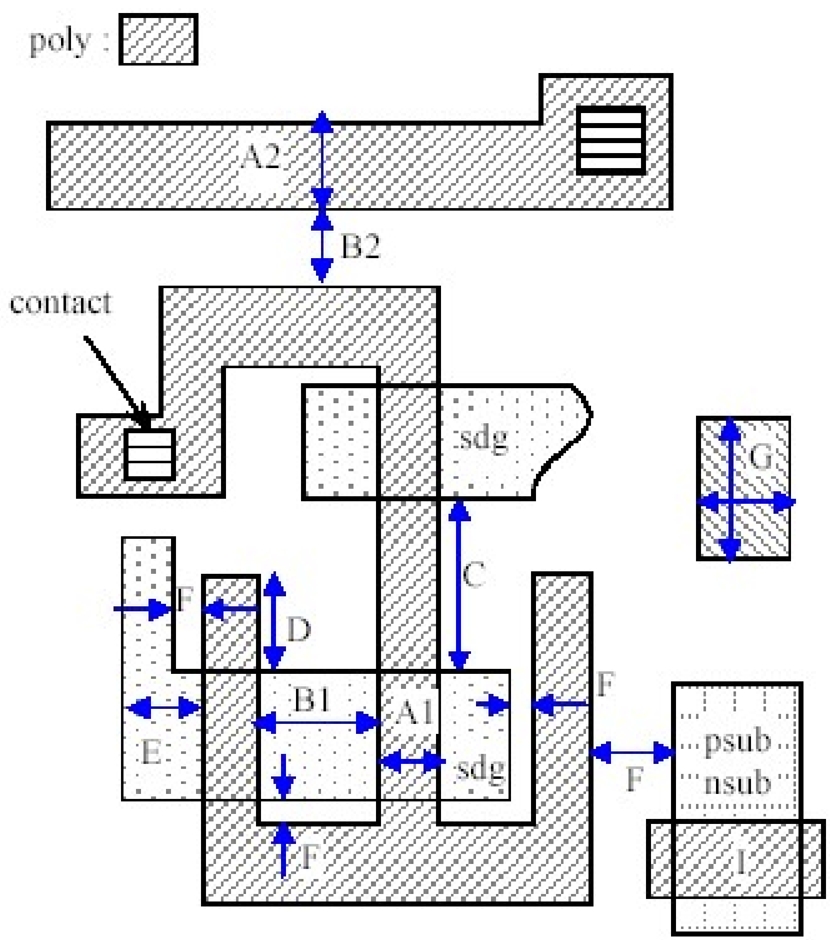
Active (sdg, psub, nsub)



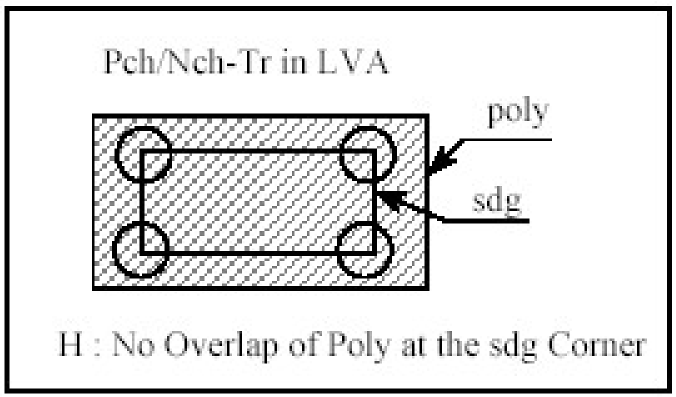
Active (sdg, psub, nsub)

序号	描述	限制
L2-A1	sdg width	0.32
L2-A2	[psub, nsub] width	0.44
L2-A3	Max. short side length of [sdg, psub, nsub]	90.00
L2-B1	sdg(N+) ~ sdg(N+) spacing, sdg(P+) ~ sdg(P+) spacing	0.32
L2-B2	sdg(N+)~nsub spacing, sdg(P+)~psub spacing	0.32
L2-C1	sdg(N+) ~ psub spacing, sdg(P+)~nsub spacing	0.40
L2-C2	nsub~psub spacing	0.40
L2-C3	Max. spacing of [sdg, psub, nsub]	10.00
L2-D	Min. [sdg, psub, nsub] area	0.313(um^2)
L2-E	Min. Dimension surrounded by [sdg, psub, nsub]	0.55×0.55

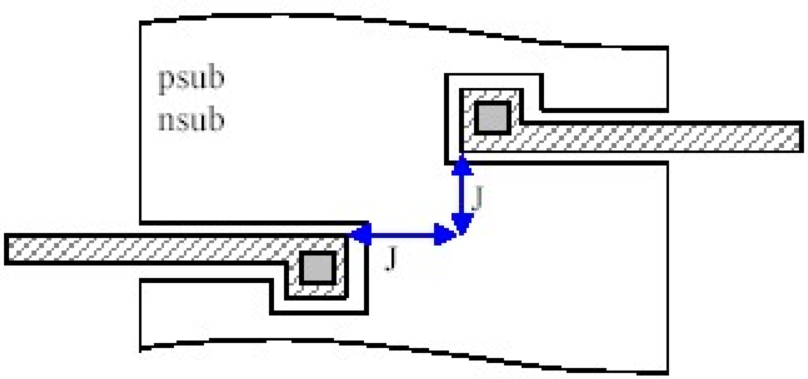
Poly



poly :
A2
B2
contact
sdg
G
C
F
D
B1
A1
E
sdg
F
psub
nsub
F
I



Pch/Nch-Tr in LVA
poly
sdg
H : No Overlap of Poly at the sdg Corner

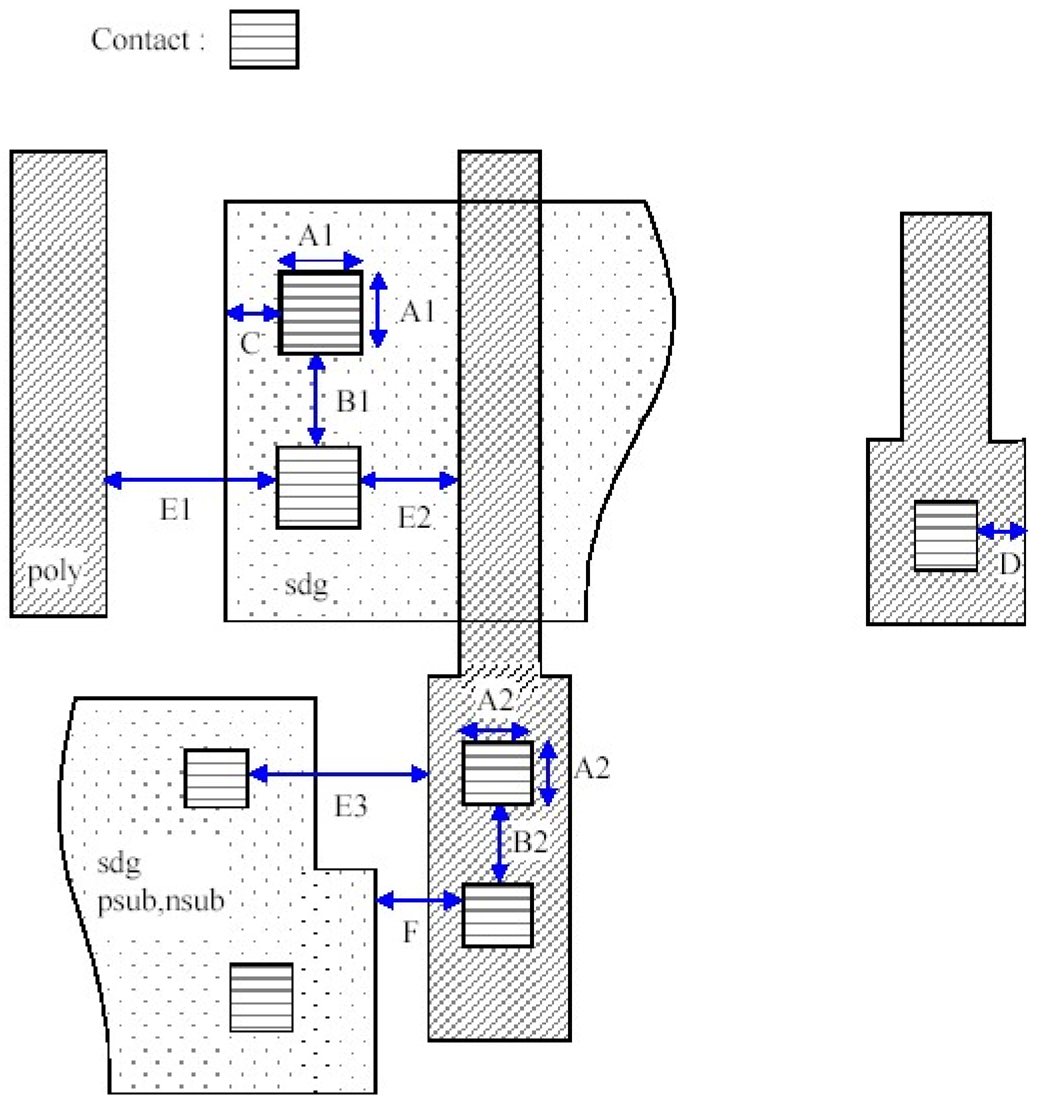


psub
nsub
J
J

Poly

序号	描述	限制
L3-A1	Poly (gate portion) width	0.32
L3-A2	Poly (interconnection portion) width	0.32
L3-B1	Poly (gate portion) spacing	0.40
L3-B2	Poly (interconnection portion) spacing	0.32
L3-C	sdg spacing with connected gate poly	0.40
L3-D	Gate portion poly fringe length	0.36
L3-E	sdg ~ poly (gate portion) length	0.44
L3-F	poly ~ [sdg, psub, nsub] spacing	0.14
L3-G	Min. poly area	0.313(um^2)
L3-H	Do not overlap poly on sdg corner portion	
L3-I	Do not overlap on psub and nsub	
L3-J	Spacing of poly surrounded by psub and nsub (in both X and Y directions)	0. 86

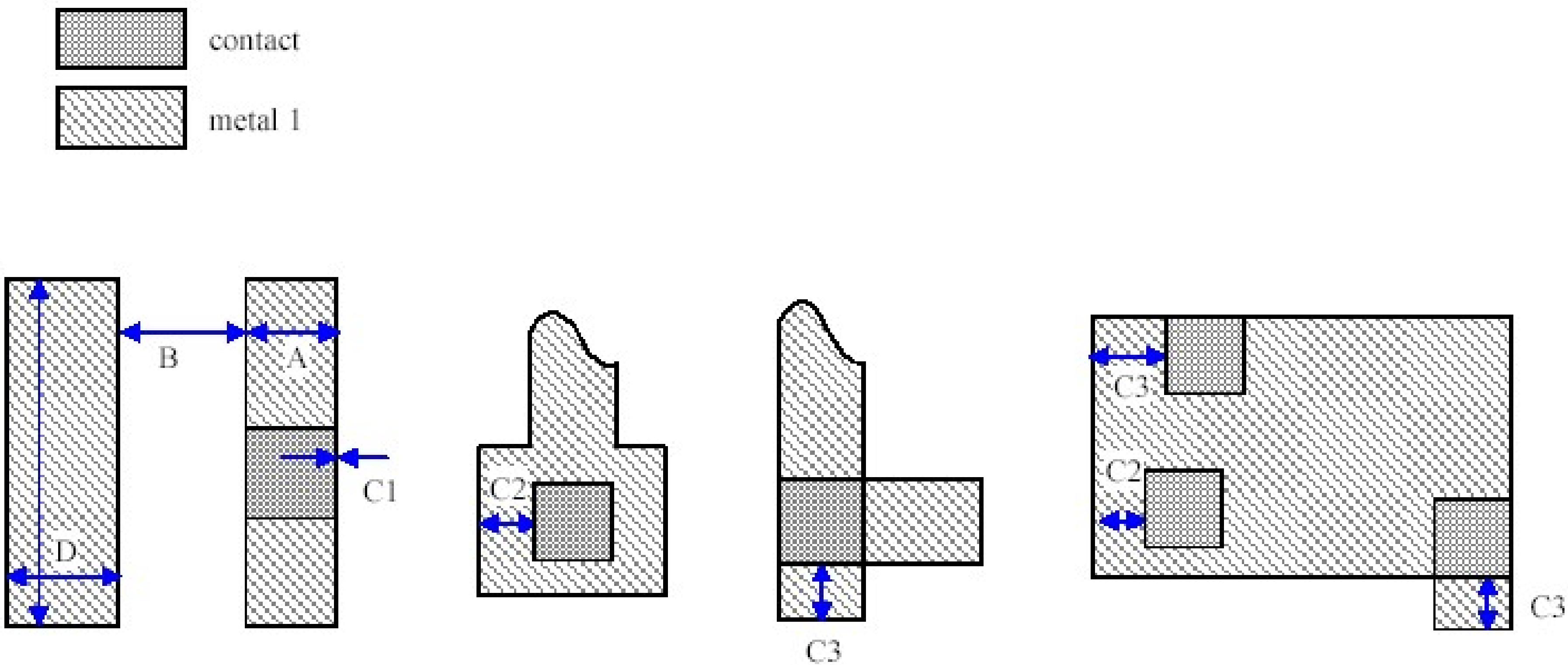
Contact



Contact

序号	描述	限制
L4-A1	Contact size on sdg, psub and nsub	0.32(Only)
L4-A2	Contact size on poly	0.32(Only)
L4-B1	Spacing of contact on sdg, psub and nsub	0.32
L4-B2	Spacing of contact on poly	0.32
L4-C	[sdg, psub, nsub] ~ contact margin	0.12
L4-D	Poly ~ contact margin	0.12
L4-E1	Spacing of contact (on sdg, psub and nsub) to poly (interconnection)	0.24
L4-E2	Spacing of contact (on sdg, psub and nsub) to poly (gate portion)	0.24
L4-E3	Spacing of contact (on sdg, psub and nsub) to poly (contact provided)	0.24
L4-F	Spacing of contact (on poly) to [sdg, psub, nsub]	0.24

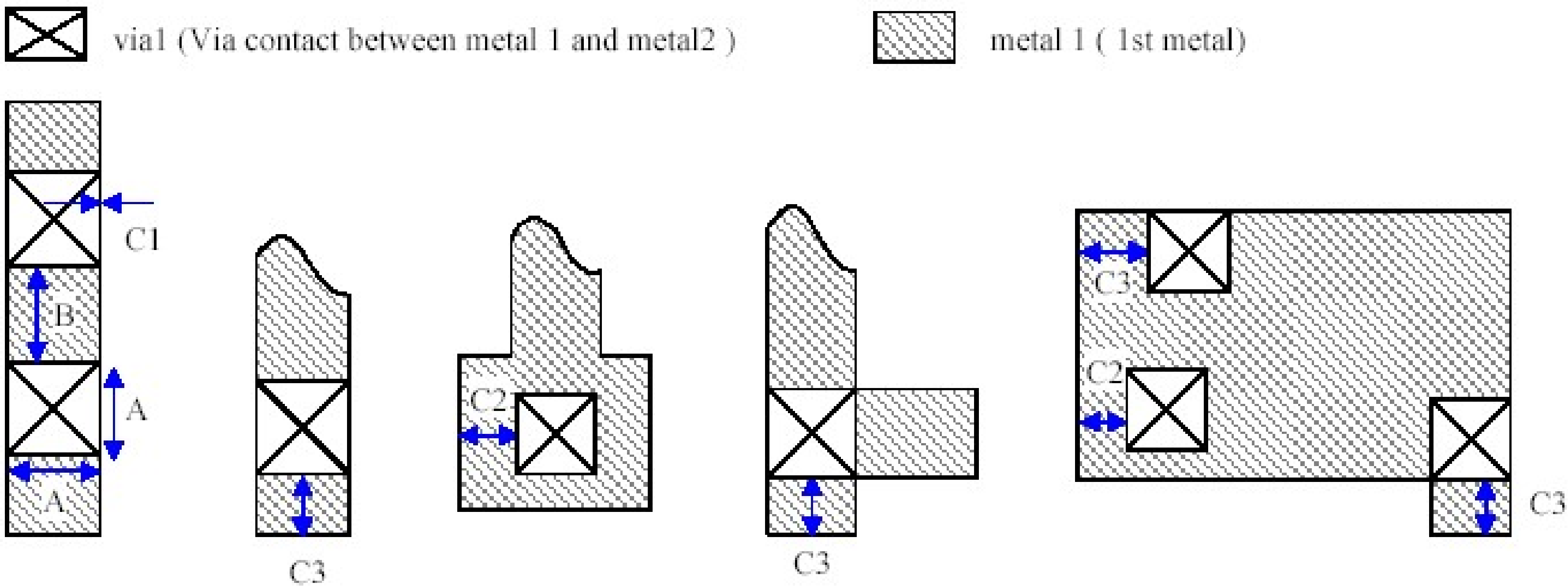
Metal 1



Metal 1

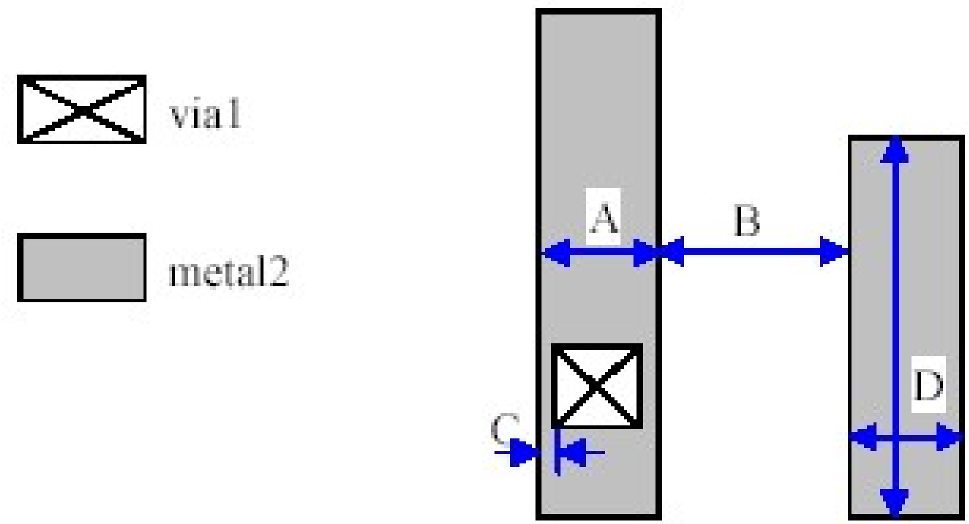
序号	描述	限制
L5-A	Min. metal 1 width	0.32
	Max. metal 1 width	50.0
L5-B	Metal 1 spacing in case of both metal width ≤ 5.0um	0.32
	Metal 1 spacing in case of any metal width ≥ 5.0um	0.48
L5-C1	Metal 1 ~ contact margin, in the straight line pattern	0.00
L5-C2	Metal 1 ~ contact margin, in the coner region	0.05
L5-C3	Metal 1 ~ contact margin, in the end overhang pattern	0.175
L5-D	Min. metal 1 area	0.2275 um^2

Via 1



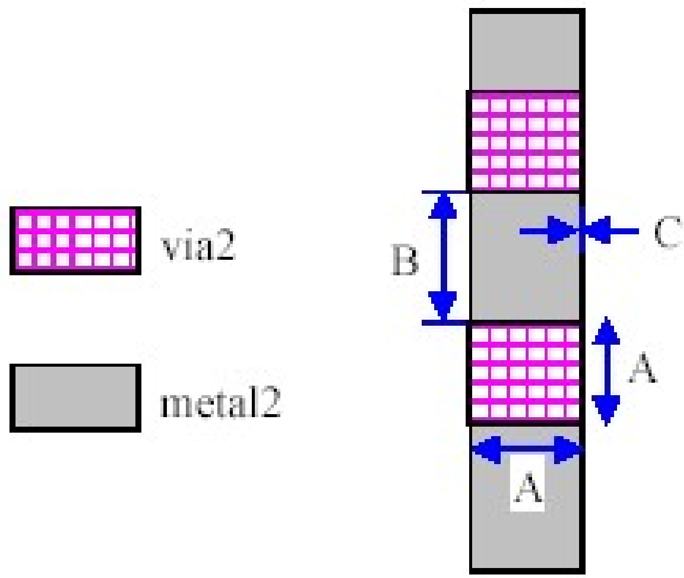
序号	描述	限制
L6-A	**Via 1 size**	**0.32(only)**
L6-B	**Via 2 spacing**	**0.325**
L6-C1	**Metal 1 ~ via1 margin, in the straight line pattern**	**0.00**
L6-C2	**Metal 1 ~ via1 margin, in the coner region**	**0.05**
L6-C3	**Metal 1 ~ via1 margin, in the end overhang pattern**	**0.175**

Metal 2



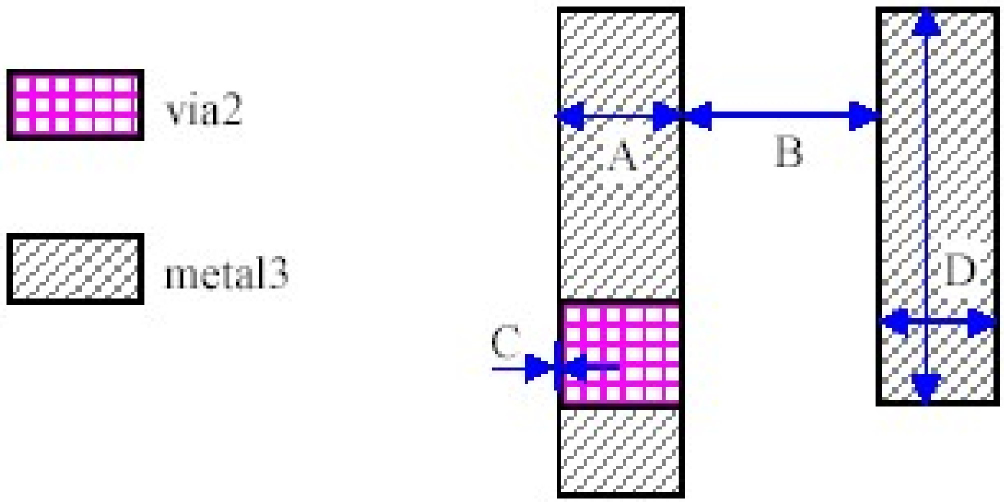
序号	描述	限制
L7-A	Min. metal2 width	0.40
	Max. metal2 width	50.00
L7-B	Metal2 spacing in case of both metal width ≤ 5um	0.40
	Metal2 spacing in case of any metal width ≥ 5um	0.64
L7-C	Metal 2 ~ via1 margin	0.05
L7-D	Min. metal2 area	$0.4275um^2$

Via 2



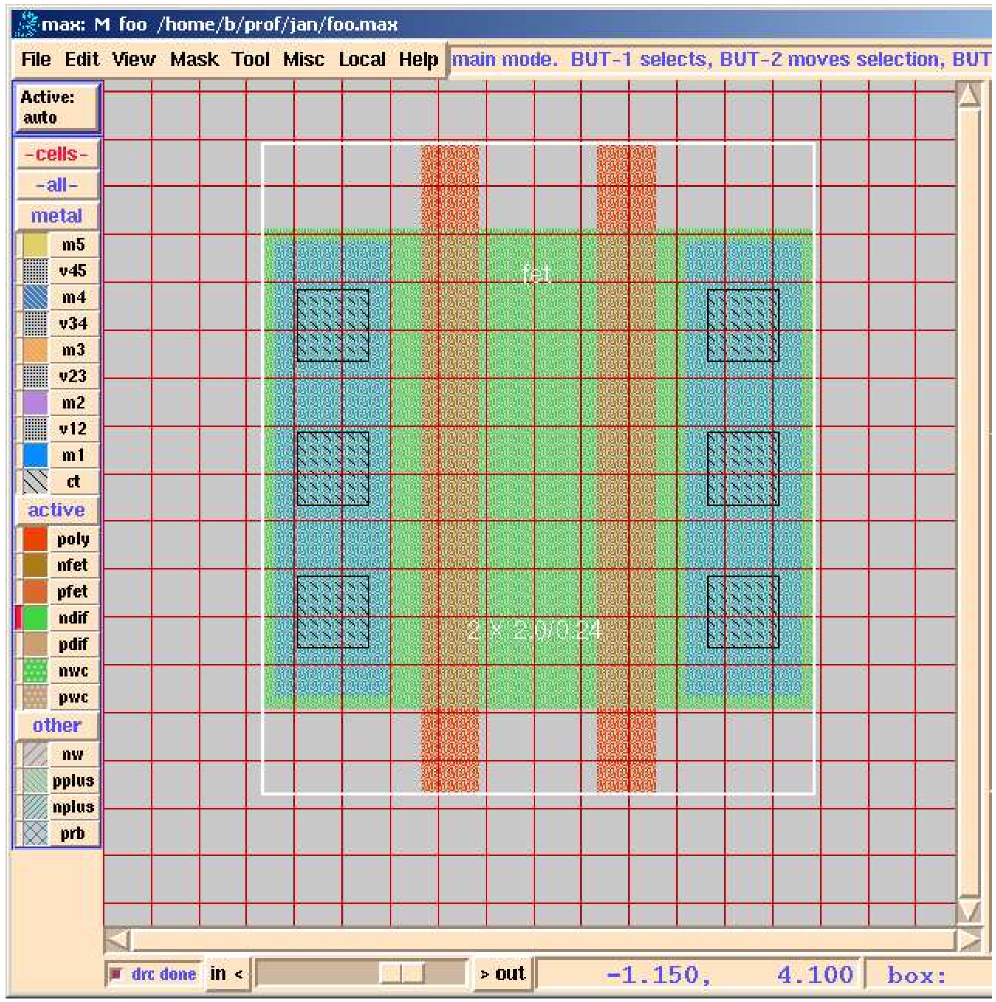
序号	描述	限制
L8-A	Via 2 size	0.40(only)
L8-B	Via 2 spacing	0.40
L8-C	Metal 2 ~ via2 margin	0.00

Metal 3

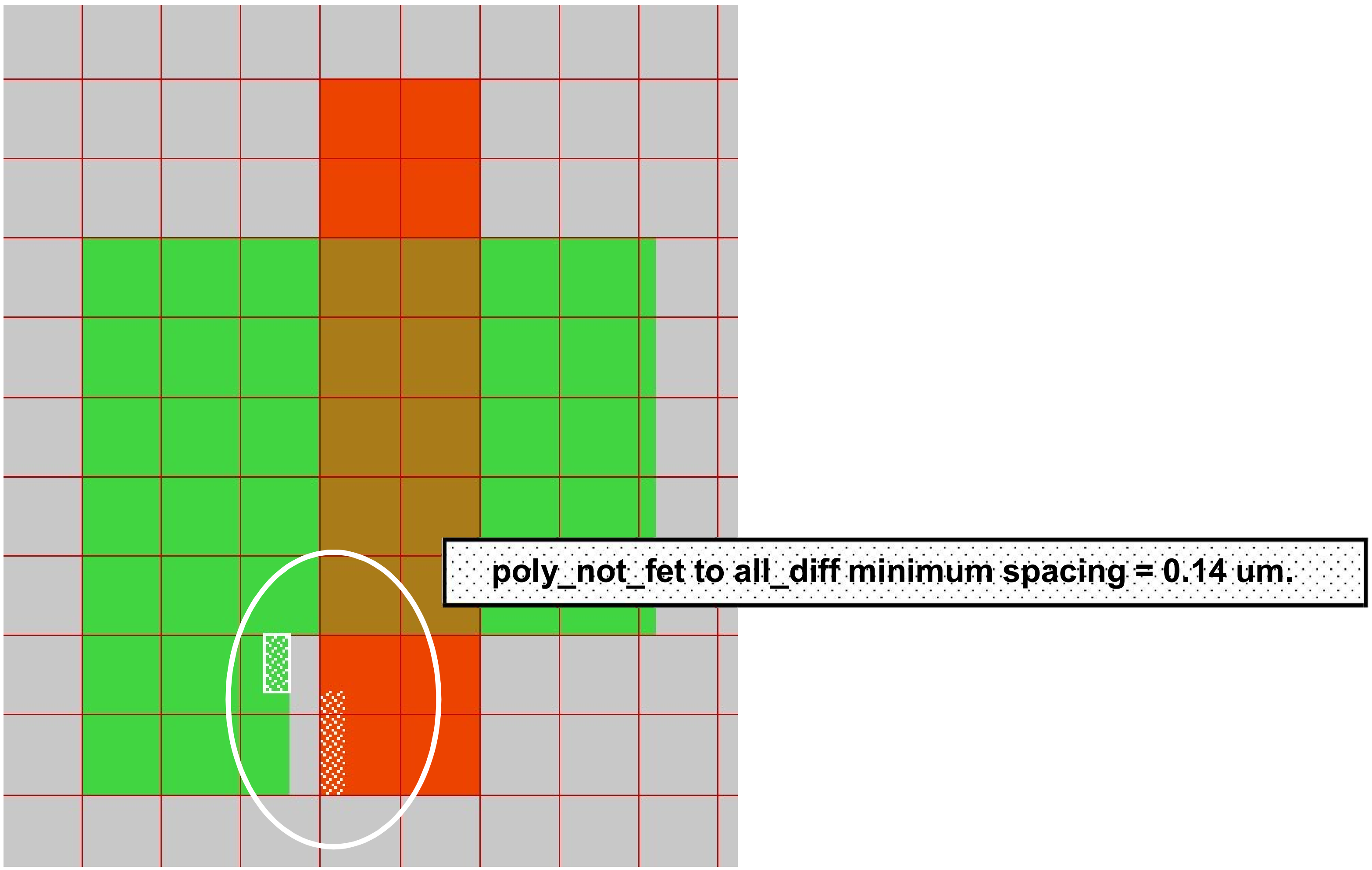


序号	描述	限制
L9-A	Min. metal3 width	0.40
	Max. metal3 width	50.00
L9-B	Metal2 spacing in case of both metal width ≤ 5um	0.40
	Metal2 spacing in case of any metal width ≥ 5um	0.64
L7-C	Metal 2 ~ via2 margin, in the straight line pattern	0.00
L7-D	Min. metal3 area	0.405um^2

版图编辑器（Layout Editor）



设计规则检查(DRC, Design Rule Checker)



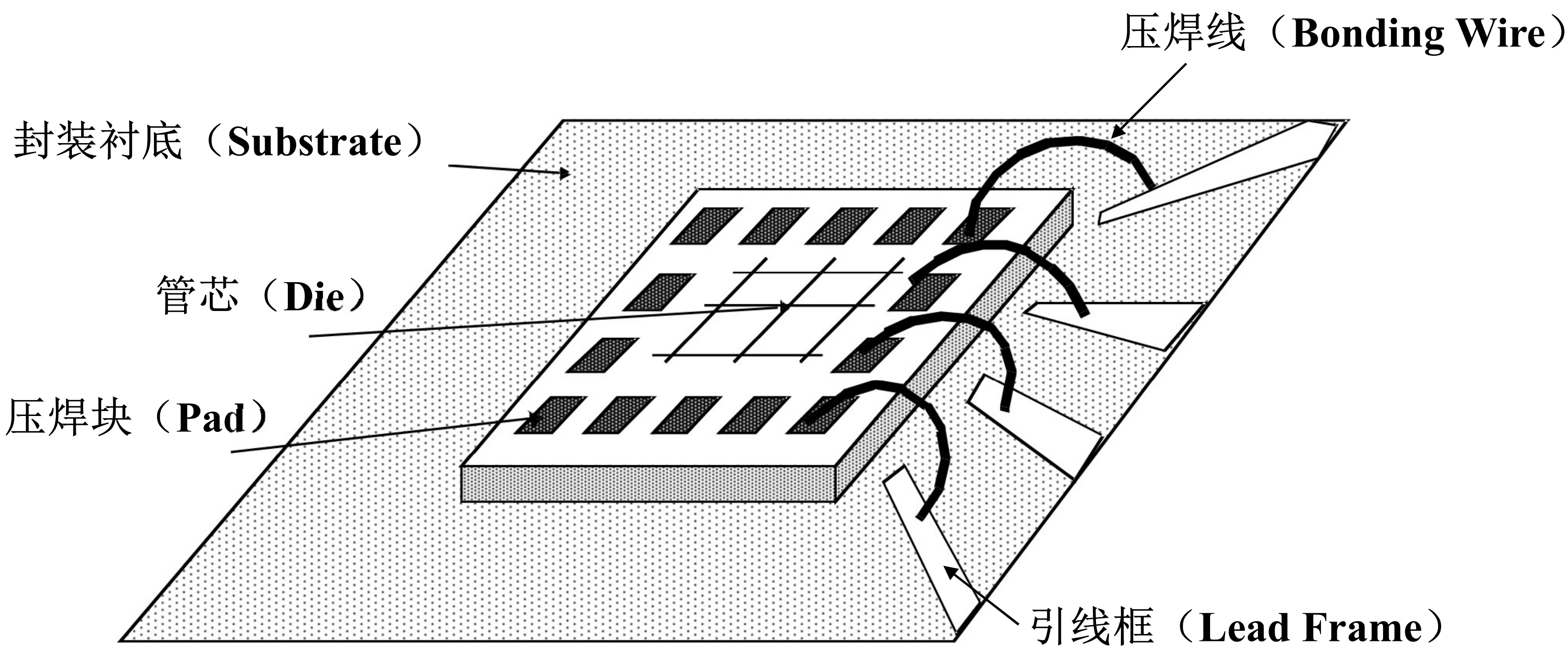
三、封装（Packaging）

封装的需求：

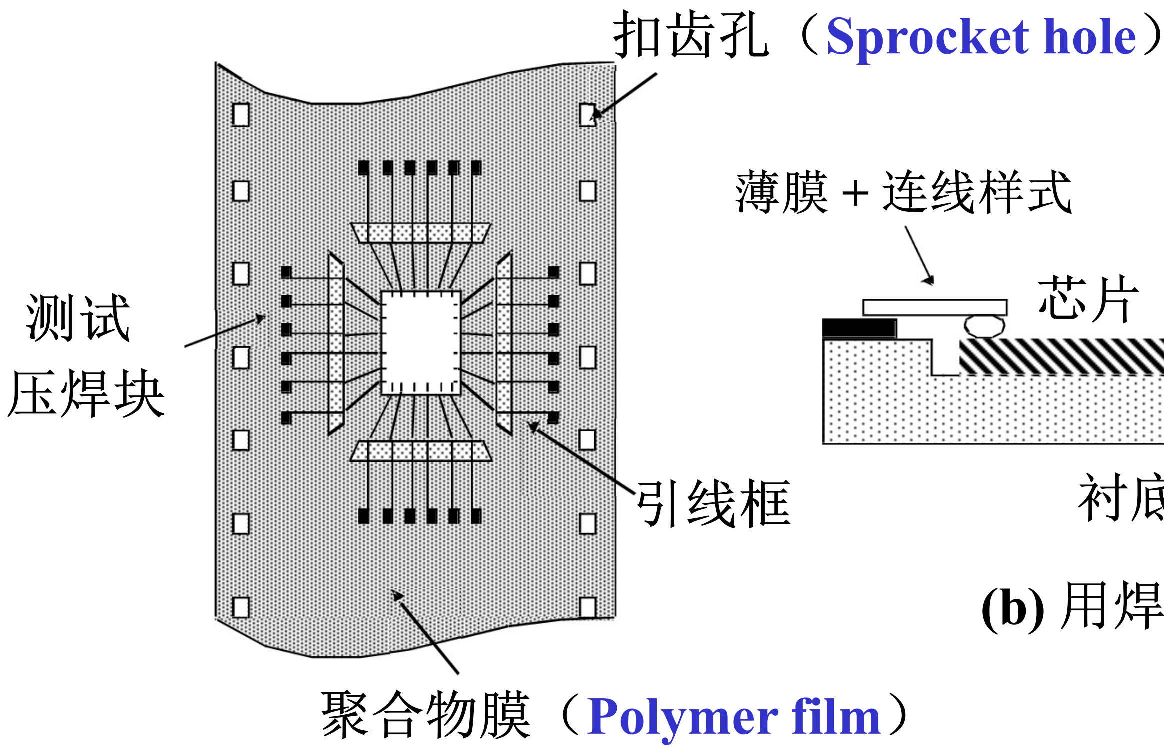
- 电气要求：低寄生参数（电阻、电容、电感）
- 机械特性：可靠和稳定
- 热学特性： 有效的散热
- 经济：低成本

压焊技术（Bonding Techniques）

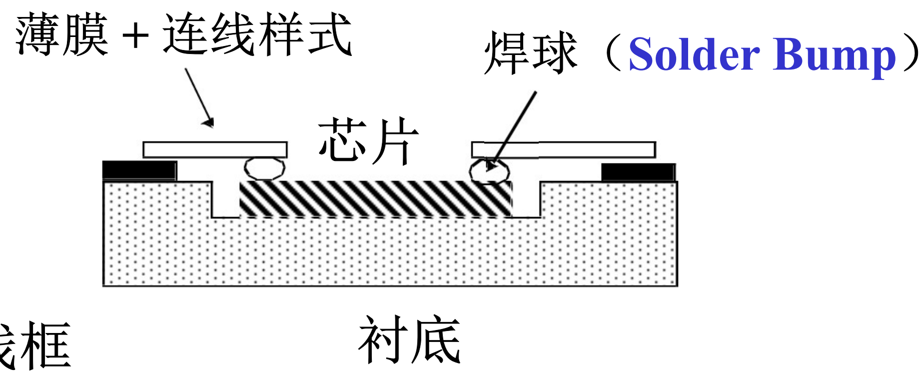
导线压焊（Wire Bonding）



载带自动压焊 (TAB, Tape-Automated Bonding)

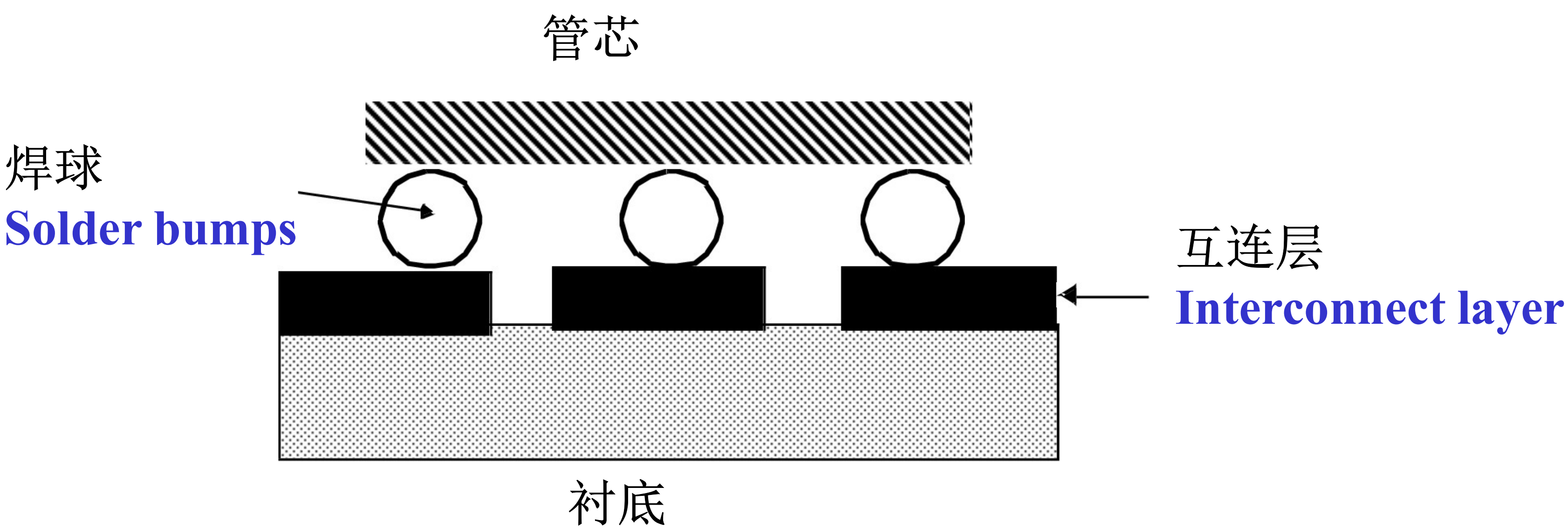


(a)带印制连线样式的聚合物载带

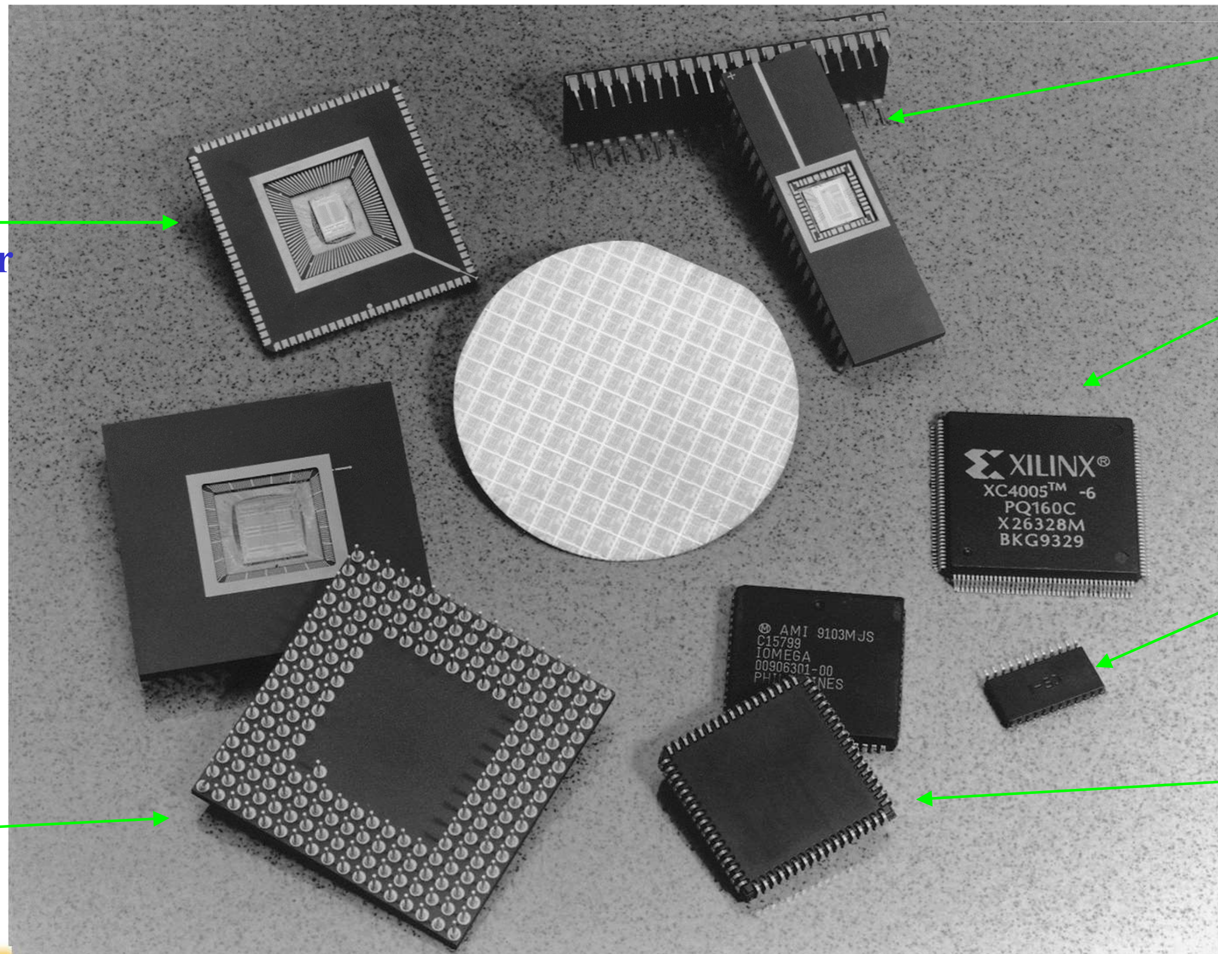


(b) 用焊球固定芯片

倒装焊（Flip-Chip Bonding）



常用封装的类型



无引线载体
Leadless Carrier
（LCC）

针栅阵列
Pin Grid Array
（PGA）

双列直插
Dual in Line
（DIP）

方型扁平封装 Quad Flat
（QFP）

小外廓IC
Small-Outline
（SOIC）

引线塑封芯片载体
Plastic Leaded
（PLCC）

封装参数

Package Type	Capacitance (pF)	Inductance (nH)
68 Pin Plastic DIP	4	35
68 Pin Ceramic DIP	7	20
256 Pin Pin Grid Array	5	15
Wire Bond	1	1
Solder Bump	0.5	0.1

Typical Capacitances and Inductances of Various Package and Bonding Styles (from [Sze83])